

Monotonic Math: A Provably Floor-Preserving Tokenomic Primitive for Revenue-Bearing On-Chain Assets

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Abstract

We present *Monotonic Math* (M^2), a tokenomic primitive that combines a fixed-supply redemption-backed token, an asymmetric-fee automated market maker, and a hardcoded revenue router to enforce a contractually monotone non-decreasing redemption floor (conditional on contract correctness and backing-stablecoin solvency). Under every protocol-defined operation—buy, sell, redeem, transfer, revenue distribution, and fee collection—the per-token claim on a stablecoin reserve is provably non-decreasing. We prove this property by case analysis over the protocol’s operation set, and derive a closed-form saturation quantity above which the optimal dump-portion of any *simultaneous* composite attacker strategy is bounded by a state-only constant independent of the attacker’s holdings—a Diamond–Dybvig analogue with reversed conclusion: later redeemers weakly dominate earlier ones, and the adversarial premium over pure redemption does not grow with holdings within the simultaneous-strategy class. We characterize the protocol’s operating envelope through closed-form deterministic projections and stochastic-revenue sensitivity bands over the four free design parameters, with a Hardhat v3 invariant-test cross-track wired in CI and a full Monte Carlo against organic-flow processes scheduled for the empirical follow-up. The mechanism is best understood as the composition of three TradFi instruments that do not coexist in any single offering we are aware of: a closed-end fund with NAV-redemption, a sponsor-funded perpetual buyback, and a Pigovian sell tax recycled into common-pool value. We position the contribution as both a DeFi mechanism-design primitive and a bridge for TradFi capital-return literature into on-chain settings.

Keywords: tokenomics, decentralized finance, mechanism design, automated market makers, redemption mechanics, capital returns, closed-end funds.

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1 Introduction

1.1 Capital Returns: From Equities to Tokens

Modern public equities have evolved a disciplined vocabulary for returning capital to holders. Dividend policy, open-market share repurchases, special distributions, and the legal apparatus of par-value redemption together constitute a century of accumulated institutional design [Grullon and Michaely, 2002, Skinner, 2008]. Each instrument resolves a different commitment problem: dividends signal earnings persistence at the cost of inflexibility; buybacks substitute for dividends

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with greater discretion but a weaker commitment [Ikenberry et al., 1995]; retractable bonds give the holder an exit at par price but at the cost of issuer optionality [Brennan and Schwartz, 1977]; and closed-end mutual funds promise net asset value (NAV) but do not enforce it on the secondary market, generating a persistent discount that has resisted decades of theoretical attention [Lee et al., 1991, Pontiff, 1996, Cherkes et al., 2009].

Tokens issued on programmable blockchains inherit none of this institutional infrastructure by default. A token’s supply, its claim on cash flows, and the conditions under which a holder can exit are whatever the issuing contract says they are. The historical record is unkind. Tokens marketed as “backed” have been backed by promises rather than redeemable claims; “floor prices” have been rhetorical descriptions of treasury-to-supply ratios rather than enforceable contracts; treasuries advertised as protocol-owned have been emptied by governance votes; and reserve currencies have spiraled into reflexive collapse under coordinated pressure [Liu et al., 2023, Klages-Mundt and Minca, 2021]. The pattern is consistent: the language of TradFi capital returns has been used without the underlying enforcement primitives that made those returns credible in the first place.

This paper takes the inverse view. We ask: what is the simplest on-chain primitive that delivers a capital-return commitment with the same enforcement discipline that TradFi instruments achieve through securities law, board fiduciary duty, and corporate-governance structures—but expressed entirely as smart-contract invariants? We give an answer, and prove it.

1.2 Monotonic Math

We present *Monotonic Math* (henceforth M^2), a tokenomic primitive composed of four immutable contracts: a fixed-supply ERC-20 token with a redeem function, a passive stablecoin treasury with no admin withdraw path, an asymmetric-fee automated market maker (AMM) implemented as a Uniswap v4 hook [Adams et al., 2024], and a revenue router with a hardcoded split. The protocol is deployed once, accepts business revenue in a backing stablecoin, and operates without governance, upgrade, or parameter-change paths.

The mechanism’s load-bearing property is what we call *floor monotonicity*: under every operation that the protocol permits—revenue deposit, buy-and-burn, redemption, transfer, liquidity-pool swap (in either direction), and fee collection—the per-token redeemable claim on the treasury, $F = T/S$, is non-decreasing. We prove this by case analysis over the protocol’s operation set (Section 4). The property is structural rather than empirical: it does not depend on market behavior, holder coordination, or operator discretion. It depends on the contract code and on the backing stablecoin remaining solvent.

Two consequences distinguish M^2 from prior on-chain primitives that have used the language of floors and redemptions. First, holders have two exit paths—redeem at the contractual floor or sell to the AMM—and *neither is floor-negative*. Redemption preserves the floor exactly (Case 3); selling to the AMM applies a Pigovian fee that, on subsequent realization, raises the floor (Cases 5 and 7). There is no protocol-defined operation that lowers the floor. Second, the asymmetric AMM fee (0.10% on buys, 3.00% on sells, with the sell fee accruing as protocol-realizable fee growth) routes adversarial pressure into common-pool value: an attempted bank run against the AMM raises the floor for those who do not run.

The mechanism is best understood, in TradFi vocabulary, as the *composition* of three instruments that do not coexist in any single offering: a closed-end fund with continuous NAV-redemption (so the secondary-market discount is bounded near zero by arbitrage between the AMM spot price and the `redeem` entry point, modulo gas, MEV, and cooldown frictions), a sponsor-funded perpetual buyback program with no announcement-execution gap (so capital returns are deterministic from earnings), and a Pigovian sell tax recycled into common-pool value rather than to a third-party fisc. No existing TradFi instrument we are aware of binds these three together—not because they could not be combined in principle, but because the

contract-as-trust substrate that enforces their irrevocable composition is absent in conventional securities law.

1.3 Contributions

We make three contributions, stated precisely so that what we claim is separated from what we do not.

C1 (Primitive). We specify M^2 as a complete, deployable protocol with four immutable contracts and quantify its design space through four free parameters: revenue rate, liquidity-pool seed fraction, asymmetric fee asymmetry, and treasury seed. We identify the genesis constraint $T_0/S_0 = L_{s,0}/L_{t,0}$ under which floor and spot coincide at launch, and characterize the protocol’s terminal state under deterministic buy-and-burn (the liquidity-pool half-life is $L_{s,0}/B$, where B is the monthly buy-and-burn size).

C2 (Proofs). Three results form the headline of our formal contribution. First, *floor monotonicity* (Theorem 2): under every operation in the protocol’s complete operation set, the per-token redemption floor $F = T/S$ is non-decreasing; the proof is by case analysis over the operation set, sharpened by an integer-arithmetic refinement showing that on-chain truncation residuals accumulate strictly in the treasury’s favor (Lemma 1). Second, a *composite-strategy bank-run theorem* (Theorem 6): within the class of *simultaneous* attacker strategies—those in which the attacker’s LP dump and treasury redemption occur in a single block without an intervening `collectFees` realization—the optimal dump quantity saturates at a closed-form quantity A^* that depends only on the AMM state and the floor, not on the attacker’s holdings N ; the attacker’s profit premium over pure redemption is then a state-only constant Δ^* , and the marginal value per held token converges to the floor as $N \rightarrow \infty$. Third, *redemption fairness* (Theorem 4): under the operations available to a holder, the per-token payout of a later redeemer weakly dominates that of an earlier redeemer, structurally inverting the open-end mutual-fund first-mover advantage of Chen et al. [2010] that underlies classical Diamond–Dybvig run dynamics [Diamond and Dybvig, 1983]. Auxiliary results bound LP exhaustion (Theorem 8), the calling equilibrium for permissionless fee collection (Theorem 9), MEV leakage under the protocol’s mitigations (Theorem 10), and spot–floor convergence under operator-revenue cessation (Theorem 7); a directional inversion of the Loss-Versus-Rebalancing identity of Milionis et al. [2024] is stated as Conjecture 5.

C3 (Quantitative envelope). We characterize the protocol’s operating envelope through Monte Carlo simulation across six scenarios (baseline, revenue-stop, mass-redemption, mass-dump, mixed-adversarial, organic-volume regimes) and sensitivity sweeps over the four free parameters. We identify Pareto frontiers between liquidity-pool longevity and floor compounding, and report stress trajectories under each adversarial composition. The simulation is validated against a Hardhat v3 invariant-test suite (Solidity `.t.sol` tests using `forge-std`-style assertions, run by Hardhat v3’s EDR rather than by Foundry) that asserts floor monotonicity on the actual compiled bytecode of the token, treasury, and revenue router; V4-mediated cases (LPBuy, LPSell, CollectFees) are exercised against the deployed mainnet PoolManager only on the `v0.1.1-paper-v1` mainnet-fork lane, and against semantics-equivalent mocks in the fast invariant lane. The two tracks must produce tolerance-band-agreeing trajectories (`ABS_TOL = 1e-3` in fee-free stable units, `REL_TOL = 1e-9`) on every shared scenario, with divergence treated as a paper-stopping bug.

What we do not claim. We do not claim spot-price monotonicity: the AMM spot can and will fall toward the floor in conditions of weak demand. We do not claim immunity to backing-stablecoin failure: the protocol holds a single deployer-chosen USD-pegged ERC-20, and its issuer’s solvency is exogenous. We do not claim robustness to operator fraud at the revenue-deposit layer (the floor is preserved even at zero revenue, but its *growth* depends on an operator continuing to route revenue; the threat-model boundary and real-world mitigations

are catalogued in Sections 3.8 and 8.3). We do not claim end-to-end mechanized verification (we sketch the path and defer to follow-up). We do not predict token returns. We do not offer regulatory classification.

1.4 Roadmap

Section 2 situates M^2 in three literatures: TradFi capital returns and closed-end-fund redemption mechanics, DeFi tokenomic protocols (with attention to OHM [Olympus DAO, 2021], RAI [Reflexer Finance, 2021], Frax [Kazemian et al., 2021], Liquity [Lauko and Pardoe, 2021], and MakerDAO [MakerDAO, 2020]), and formal-methods work on automated market makers and on-chain mechanism design. Section 3 specifies the four contracts and introduces a threat model. Section 4 states and proves the floor-monotonicity, redemption-fairness, and integer-arithmetic redemption results. Section 5 formalizes the bank-run game and contrasts it with the Terra–Luna collapse [Liu et al., 2023]. Section 6 reports the simulation. Section 7 provides a cross-protocol comparison matrix. Section 8 catalogs limitations, open mathematical problems, and an academic regulatory framing. Section 9 closes.

2 Background and Related Work

We position M^2 at the intersection of four literatures: TradFi capital-return mechanisms, DeFi tokenomic protocols, formal methods and mechanism design for automated market makers, and the economics of stablecoin pegs and runs. Each literature contributes a vocabulary and a set of negative results that the M^2 design responds to.

2.1 TradFi Capital-Return Mechanisms

Three families of TradFi instruments inform the M^2 design directly.

Buybacks and dividend policy. Share repurchases have substituted for cash dividends as the dominant capital-return channel in U.S. equities since the late 1980s [Grullon and Michaely, 2002]. The substitution is partial: buybacks track earnings more flexibly than dividends [Skinner, 2008], but that flexibility is precisely what weakens their commitment value, and Ikenberry et al. [1995] document a resulting market underreaction to repurchase announcements. M^2 translates the buyback mechanic into an automated and deterministic form: a fixed 50% slice of each revenue inflow is routed to an on-chain market buy whose execution is enforced by the protocol contract rather than by board discretion. There is no announcement-execution gap. Furthermore, by burning rather than retiring to treasury, the protocol guarantees that bought-back tokens cannot be re-issued—a guarantee that no public-equity buyback program can credibly make.

Closed-end funds and NAV redemption. The persistent discount-to-NAV of closed-end funds (CEFs) is the canonical asset-pricing anomaly that maps most directly to M^2 [Lee et al., 1991, Pontiff, 1996]. CEFs publish a net asset value but provide no redemption mechanism: investors who want out must sell on the secondary market, where the discount can persist for decades. Cherkes et al. [2009] model the discount as a liquidity premium; Berk and Stanton [2007] attribute part of it to management-fee capitalization. M^2 is structurally a CEF whose NAV-redemption mechanism is enforced at the contract level: the discount cannot persist because any rational arbitrageur faced with $P < F$ buys on the AMM and redeems against the treasury. What M^2 adds beyond NAV-redemption is *monotone NAV*—in a CEF, the NAV itself fluctuates with portfolio mark-to-market, whereas in M^2 , the contractually-defined $F = T/S$ is non-decreasing by Theorem 2.

Retractable bonds and other put options. Brennan and Schwartz [1977] analyzed retractable bonds—fixed-income instruments that grant the holder a put option to redeem at par. The optionality is analytically equivalent to M^2 ’s `redeem` function: the holder may exit at a known price independent of secondary-market conditions. The differences are that retractable bonds redeem at a fixed schedule of par values, whereas M^2 redeems at a state-dependent (but monotonically rising) floor, and that bond covenants can be modified by majority of holders whereas M^2 ’s redemption is immutable.

Mutual-fund redemption fragility. Chen et al. [2010] document strategic complementarities in open-end mutual fund redemptions: when illiquid-asset funds face redemption pressure, early redeemers extract value at the expense of remaining holders, generating a first-mover advantage that can produce run dynamics. M^2 ’s Theorem 4 (Section 4) is precisely the negation: redemption preserves the floor exactly, so there is no first-mover advantage in floor terms. The classical Diamond and Dybvig [1983] bank-run equilibrium fails to obtain in M^2 for the same structural reason.

Other reference instruments. REIT mandatory distributions (the U.S. Internal Revenue Code §856 requires REITs to distribute $\geq 90\%$ of taxable income to retain pass-through status; see Hartzell et al. [2014] for a study of institutional governance in this setting) provide a TradFi analogue to M^2 ’s hardcoded 50/50 revenue split, with the difference that REIT distributions can be retained at the cost of losing tax-advantaged status, whereas M^2 ’s split is a contract constant. Dividend reinvestment plans [Scholes and Wolfson, 1989] compound returns into share accumulation, in contrast to M^2 ’s compounding into per-token floor. Money-market funds redeeming at par [Kacperczyk and Schnabl, 2013] provide the closest peg analogue, though M^2 has no peg and instead has a monotone floor.

Synthesis. Table 1 summarizes the mapping. No single TradFi instrument composes the three features that M^2 composes (the headline framing in Section 1 names them); the composition is enabled by the contract-as-trust substrate rather than by any individual feature being novel.

2.2 DeFi Tokenomic Protocols

Werner et al. [2022] provide a taxonomic survey of decentralized finance that frames the landscape we situate M^2 within. We comment specifically on prior protocols that have engaged with redemption, treasury backing, or perpetual buybacks.

OlympusDAO and forks. OHM [Olympus DAO, 2021] introduced the “backed not pegged” framing and the concept of risk-free-value (RFV) per token. The framing was rhetorical rather than enforced: holders could not redeem OHM against the RFV treasury, and treasury composition could be changed by governance vote. The empirical record of OHM forks (Wonderland, Klima, and others) is dominated by treasury raids and confidence collapses. M^2 responds to this lineage by (i) making redemption a first-class contract function, (ii) removing governance entirely, and (iii) holding only a single transparent stablecoin in treasury. Crucially, M^2 ’s supply only decreases over time, in contrast to OHM’s staking-rebase inflation.

Reflexer Finance / RAI. Reflexer Finance [2021] introduced a PID-controlled redemption price that can diverge from market price. RAI and M^2 share the conceptual move of separating “redemption price” from “market price,” but RAI’s redemption price is mean-reverting around a control target whereas M^2 ’s is monotone-nondecreasing.

Table 1: TradFi capital-return instruments and their M² analogues.

TradFi instrument	Key feature	M ² analogue	What M ² adds
Closed-end fund	NAV is published but not enforced; persistent discount	Floor $F = T/S$ enforced by redeem	NAV is monotone-nondecreasing
Open-market buyback	Sponsor-discretionary; announcement-execution gap	Hardcoded 50% revenue split to buy-and-burn	No discretion; bought-back tokens are burned, not retired
Retractable bond	Holder put at par on schedule	redeem at floor, on demand	State-dependent rising floor; immutable terms
DRIP	Compounding via share accumulation	Compounding via floor growth	Holder needs no action; compounding is automatic
REIT distribution	Required by tax structure	Required by contract constant	Distribution enforced by an on-chain redemption primitive, not a corporate intermediary (deployment-jurisdiction analysis required; see §8.6)
Tobin tax / Sec. 31 fee	Externality charge to state	3% sell fee accruing to floor	Pigovian charge recycled to common pool

Frax, Liquity, MakerDAO. [Kazemian et al. \[2021\]](#) introduced fractional-algorithmic stablecoins. [Lauko and Pardoe \[2021\]](#) demonstrated a governance-free, immutable, ETH-collateralized stablecoin (LUSD) and is the closest prior art for M²’s “immutable plus no governance” design stance. [MakerDAO \[2020\]](#) pioneered multi-collateral, governance-rich CDP-style redemption. Each of these systems is a stablecoin (pegged or controlled); M² is not a stablecoin. The redemption mechanism is shared in structure, but M²’s redemption is to a state-dependent floor rather than to a target peg.

Bonding curves and protocol-owned liquidity. [Hertzog et al. \[2017\]](#) introduced bonding curves as a primitive for protocol-enforced redemption pricing; M²’s $F = T/S$ is a degenerate bonding curve where the per-token payout is independent of position on the curve. [Heimbach et al. \[2021\]](#) characterize liquidity-provider entry and exit behavior in automated market makers, a structural sibling of the protocol-owned-liquidity regime M² adopts; M² inherits this lineage but locks the position permanently.

2.3 Formal Methods and Mechanism Design for AMMs

[Angeris and Chitra \[2020\]](#) provide the canonical formal treatment of constant-function market-maker properties; [Angeris et al. \[2021\]](#) extend this to no-arbitrage results for Uniswap. [Bartoletti et al. \[2021\]](#) develop a theory of automated market makers that informs the proof style we adopt in Section 4. [Adams et al. \[2024\]](#) specify the v4 hook architecture that M² uses for asymmetric per-swap fee overrides via `beforeSwap` (with `LPFeeLibrary.OVERRIDE_FEE_FLAG` on dynamic-fee pools).

On the adversarial side, [Daian et al. \[2020\]](#) introduced the modern study of maximum extractable value, and [Qin et al. \[2022\]](#) provide empirical magnitudes. [Heimbach and Wattenhofer \[2022\]](#) apply game-theoretic mechanism design to sandwich attack prevention; the principles inform M²’s anti-MEV mitigations (private orderflow, randomized delay, TWAP execution). [Babel et al. \[2023\]](#) introduce a methodology for adversary-bounded value extraction proofs in

smart-contract economic-security analysis, which is the proof genre our Section 5 bank-run analysis lives in.

A distinct strand of recent work concerns *Loss-Versus-Rebalancing* (LVR) for AMM liquidity providers [Milionis et al., 2024]. The standard LVR framing presents the LP-versus-external-CEX price differential as a cost extracted by arbitrageurs. M^2 inverts this framing structurally: the LP position is owned by the protocol, the protocol’s value flows into the floor, and—under the asymmetric fee mechanism—LVR-equivalent value is plausibly captured rather than lost. We conjecture this inversion as Conjecture 5 and discuss the unit-reconciliation gap that prevents promoting it to a theorem in Section 8.5.

For broader context on formal verification, Tolmach et al. [2022] survey smart-contract specification and verification methods. The skeptical view of decentralized market-making is articulated in Park [2023]; we engage with this critique in Section 8 given that M^2 ’s LP is internal rather than external.

2.4 Stablecoin Run Dynamics

Klages-Mundt et al. [2020] provide an economic taxonomy of stablecoin designs and risk-based models. The Terra–Luna collapse [Liu et al., 2023, Briola et al., 2023] is the benchmark case study for how endogenous-collateral algorithmic stablecoins enter a reflexive death spiral under coordinated redemption pressure. We discuss M^2 ’s structural disanalogy with Terra in Section 5: M^2 ’s collateral is exogenous (a deployer-chosen USD-pegged stablecoin) and one-way-in (the only path stable leaves the treasury is *redeem*, which is in lockstep with supply burn). The specific endogenous-collateral feedback loop that drove the Terra collapse is structurally absent from M^2 , conditional on the backing stablecoin remaining solvent and on contract correctness.

Adjacent work on peg dynamics includes Gorton et al. [2022] on leverage and stablecoin pegs, Catalini et al. [2022] on the economic foundations of stablecoins, Routledge and Zetlin-Jones [2022] on currency stability via blockchain, and d’Avernas et al. [2023] on the broader question of stablecoin stability under reserve composition. Eichengreen [2019] situates the debate historically. M^2 is explicitly not a stablecoin—it has no peg target, no e-money classification under MiCA’s tests, and a deliberately variable spot price. Nonetheless the analytical tools of the stablecoin literature transfer because the redemption mechanic poses the same liquidity questions as a depeg event.

3 Protocol Specification

This section specifies M^2 as a complete deployable system: four immutable contracts, the genesis ceremony that links them, the operations they expose, and a threat model that delimits the conditions under which the formal guarantees of Section 4 hold. We adhere to the notation introduced in Section 1 and standardized in Appendix A: T for the treasury stablecoin balance, S for the total token supply, $F = T/S$ for the floor price, (L_t, L_s) for the active reserves of the protocol-owned liquidity-pool position, and $P = L_s/L_t$ for the AMM spot price.

3.1 System overview

M^2 is composed of four contracts—a token, a treasury, an asymmetric-fee liquidity-pool hook, and a revenue router—together with a single off-protocol agent, the *operator*, who routes business revenue (denominated in the backing stablecoin) into the router. Figure 1 shows the full system. Each contract is deployed once and has no admin key, no governance hook, no upgrade path, and no parameter-change interface. There is no proxy pattern: the bytecode at each address is its final form.

The high-level dynamics are: (i) the operator deposits revenue X (denominated in the backing stable) into the router; (ii) the router atomically deposits $X/2$ into the treasury and

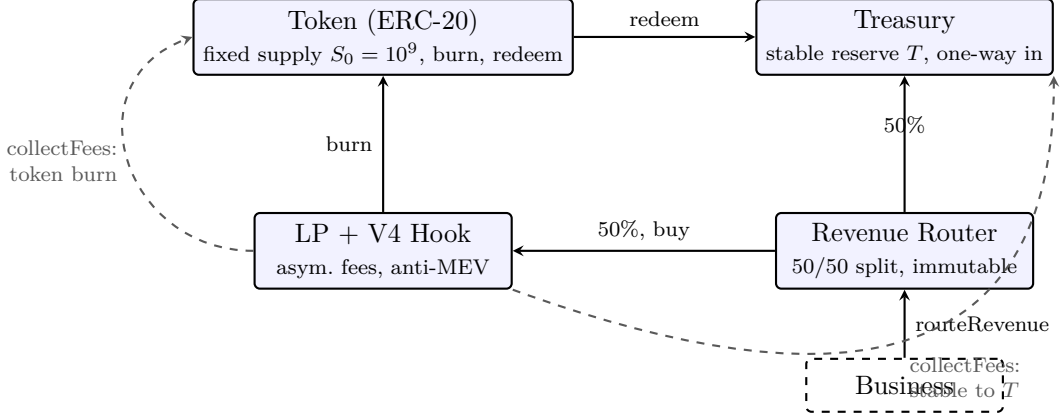


Figure 1: System architecture. Blue-tinted boxes are immutable contracts; the dashed box denotes the off-protocol operator. Solid arrows mark primary flows; dashed gray arrows mark the permissionless `collectFees` fee-realization paths (token side: 99.75% burned, 0.25% to caller; stable side: 99.75% to treasury, 0.25% to caller). All inflows to Treasury are one-way: stablecoin can leave only via `redeem`, which simultaneously burns the redeemed tokens.

uses $X/2$ to execute a market buy on the protocol-owned LP, burning the tokens received; (iii) holders may at any time invoke either `redeem` on the token (atomic, no cooldown, no fee) or a swap against the LP (with an asymmetric per-direction fee enforced by the hook); (iv) any external caller may invoke `collectFees`, which realizes the LP-position’s accrued fee growth and routes 99.75% of the token-side fees to burn and 99.75% of the stable-side fees to the treasury, with the residual 0.25% per side paid as a bounty to the caller.

3.2 Token

The token is an ERC-20 contract with 18 decimals and a total supply $S_0 = 10^9$ tokens minted to the deployer in a single transaction at genesis. The `mint` function is gated by an immutable constructor-set sentinel that, once consumed by the genesis mint, blocks all further minting. Three roles can call `burn(address, uint256)`: the hook (when realizing token-side LP fees), the router (when executing buy-and-burn), and the token’s own `redeem` entry point. No other path reduces supply.

The `redeem(uint256 amount)` function is the load-bearing TradFi analogue. It is atomic, non-pausable, and has no cooldown, no fee, and no maximum. It burns the caller’s `amount` tokens, computes the payout as `amount × floorPrice()` in the stable’s smallest unit, and instructs the treasury to transfer that amount to the caller via the privileged `payRedemption` interface. The function reverts only if the caller’s balance is insufficient or if $S = 0$ (the boundary case is handled by a dedicated `SupplyExhausted` error; by construction, when $S = 0$ the treasury balance is also zero, so the revert locks no value).

Precision and stable-decimals neutrality. The floor function returns an 18-decimal fixed-point value regardless of the backing stable’s native decimals d_s :

$$\text{floorPrice}() = \frac{T \cdot 10^{18+18-d_s}}{S}, \quad (1)$$

with the wide intermediate $T \cdot 10^{18+18-d_s}$ taken in `uint256`. For USDC ($d_s = 6$) the upscale factor is 10^{30} ; for DAI ($d_s = 18$) it is 10^{18} . With the genesis parameters of Section 3.7, the intermediate is bounded above by $10^{18+18-6} \cdot T \leq 10^{30} \cdot 10^{12} (\$) = 10^{42} \ll 2^{256} \approx 1.16 \cdot 10^{77}$, so no overflow can occur in any realistic deployment.

On redemption, the gross payout `amount × floorPrice()` is divided down to the stable’s smallest unit before transfer. The truncation can leave at most $S - 1$ smallest-units of dust in the treasury’s favor per redemption call; Lemma 1 (Section 4) shows that this residual strictly raises the floor and never lowers it.

3.3 Treasury

The treasury is the passive custodian of the backing stablecoin. It exposes exactly one privileged entry point, `payRedemption(address user, uint256 stableAmount)`, callable only by the token contract. It has no admin role, no withdraw function, no pause switch, no upgrade path, and no governance hook. The two inflow paths—the router’s 50% split and the 99.75% stable-side share from `collectFees`—are one-way: stablecoin can leave only through `payRedemption`, and only in lockstep with a burn of the redeemed tokens via the token contract’s `redeem` flow.

The contract’s invariants are: (i) $T \geq 0$ at all times (enforced by the underlying ERC-20); (ii) every decrease of T is accompanied by a same-block burn of a token amount whose floor-product equals the stable outflow up to integer truncation in the treasury’s favor (Lemma 1); (iii) no caller other than the token contract can decrease T . Together these properties give the treasury the structural role of a CEF reserve whose NAV-redemption mechanism is enforced at the bytecode level.

Genesis seed. The deployer transfers the treasury’s genesis seed T_0 in the same transaction that deploys the system. The seed must satisfy the floor-spot constraint $T_0/S_0 = L_{s,0}/L_{t,0}$, so that $F_0 = P_0$ at launch (Section 3.7). For the canonical parameters of this paper, $T_0 = \$1,000,000$.

3.4 Liquidity pool and V4 hook

The protocol-owned LP is a Uniswap v4 pool with token currency `Token` and stable currency `Stable`, initialized with the `DYNAMIC_FEE_FLAG` so that fees are determined per-swap by the hook [Adams et al., 2024]. The protocol holds the LP position permanently; the contract owning the position has no externally callable removal interface, so the liquidity is provably unwithdrawable.

Full-range position and V2-equivalence of the constant-product algebra. The protocol-owned position is initialized as a *single full-range position* spanning the maximum admissible tick interval $[\text{MIN_TICK}, \text{MAX_TICK}]$ at the pool’s tick spacing. Under this configuration the pool has exactly one piecewise-constant liquidity segment, so v4’s *active liquidity* equals the position’s liquidity at every reachable spot price. Concretely, the per-tick bookkeeping that v4 uses to support concentrated liquidity—tick-crossing, partial-range positions, segmented liquidity profiles—does not engage at any state the protocol can reach: the active reserves (L_t, L_s) are the V2-style $x \cdot y = k$ reserves of a single pool, computed from the position’s `liquidity` and the current `sqrtpPriceX96` via the standard full-range identities. The constant-product invariant $k = L_t \cdot L_s$ (Section 4.1, Equation (8)) is exact at every reachable state, not an approximation, and the per-swap update preserves k exactly modulo the asymmetric-fee deduction described below. V4 is retained for its per-swap hook-fee primitive (Equation (2)) and the position-locking semantics that make the LP unwithdrawable; no concentrated-liquidity feature is used. Throughout the paper, (L_t, L_s) are read as the V2-style reserves of a full-range v4 position, with the v4 fee accumulator reconciled to the raw-mass framing (Φ_t, Φ_s) in the footnote on Equation (2).

Asymmetric fee mechanism. On every swap, the hook’s `beforeSwap` callback returns a fee override (OR’d with `LPFeeLibrary.OVERRIDE_FEE_FLAG`) determined by swap direction:

$$\text{fee}(\text{direction}) = \begin{cases} 0.10\% & \text{if input is stable (buy)} \\ 3.00\% & \text{if input is token (sell)} \end{cases} \quad (2)$$

The fee is subtracted from the swap’s input amount before the constant-product curve is applied: `SwapMath.computeSwapStep` deducts `step.feeAmount` from `amountIn` and folds the deduction into the input-currency-side accumulator, which we denote by raw-token mass (Φ_t, Φ_s) in Section 4.1.¹ Because the LP position is owned by the protocol, the accrued fee growth is realizable only by the protocol itself, via a subsequent `collectFees` call (Section 3.6). The 0.10% buy fee is a design choice rather than a revenue mechanism: v4’s per-swap fee override (gated by `LPFeeLibrary.OVERRIDE_FEE_FLAG`) admits any 24-bit fee value, including zero, so a zero buy fee is technically expressible. We set the buy fee to a small nonzero value to keep both swap directions on the same fee-override code path and to standardize input-side fee accumulation across buy and sell legs; the resulting basis-point inflow to Φ_s is incidental rather than load-bearing. The 3.00% sell fee, by contrast, is load-bearing: it converts every adversarial exit on the LP into common-pool value.

Why v4 rather than v2 or v3. Uniswap v2 fees compound back into the pool’s reserves and cannot be routed; v2 also has no per-swap fee primitive, so no asymmetric fee is expressible without writing a wrapper AMM. Uniswap v3 supports routed fees via `collect()` on NFT positions but offers only fixed per-pool fee tiers. Only v4’s `beforeSwap` hook allows a per-swap, direction-conditional fee on a single pool. We accept v4’s larger verification surface (a newer and less battle-tested codebase than v2/v3) as the structural trade-off for the asymmetric-fee primitive on which Cases 5 and 7 of Theorem 2 depend.

Two exit paths, both floor-non-decreasing. A holder may exit via `redeem` at the contractually-defined floor or via a token-to-stable swap on the LP. The redemption path preserves the floor exactly (Case 3 of Theorem 2). The LP-sell path holds the floor flat at the moment of the swap and strictly raises it at the next `collectFees` realization (Cases 5 and 7). No protocol-defined operation lowers the floor.

Anti-MEV mitigations. Protocol-initiated buy-and-burn transactions are predictable in timing and direction, making them sandwichable in expectation. The hook and router together apply three mitigations: (i) private orderflow routing appropriate to the deployment chain (Flashbots Protect on Ethereum mainnet; the native sequencer-protected mempool on chains that provide one); (ii) a uniform random delay $\tau \sim U[0, 12\text{h}]$ between the monthly tick and the actual buy execution; (iii) optional TWAP execution that splits the monthly buy across N sub-buys within the delay window. Theorem 10 (Section 5) bounds the residual MEV leakage under these mitigations as a function of private-mempool inclusion probability and TWAP granularity.

3.5 Revenue router

The router is a one-function contract:

```
function routeRevenue(uint256 stableAmount) external onlyDepositor;
```

¹The v4 implementation stores the accumulator as a Q128.128 fixed-point per-liquidity quantity `feeGrowthGlobalX128`; the position’s realizable fee at collection is $K_{\text{real}} = (\Delta\text{accumulator}) \cdot \text{liquidity} / 2^{128}$, which equals the raw input-side fee mass deposited since the last collection. The raw-token framing used throughout this paper is dimensionally equivalent to the v4 accumulator and avoids the Q128.128 bookkeeping in the proofs.

On invocation, it pulls `stableAmount` from the immutably-set `depositor` address, deposits exactly half ($\lfloor \text{stableAmount}/2 \rfloor$) into the treasury, and uses the remainder ($\lceil \text{stableAmount}/2 \rceil$) to execute a buy on the LP via the hook. The tokens received from the buy are burned in the same transaction.

The split is a `constant` in the router’s bytecode—there is no setter, no upgrade path, no governance hook. A different split would require deploying a separate protocol with a separate token. The depositor address is set in the router’s constructor and is immutable; substitution after launch is impossible without redeploying the entire system.

Why a fixed 50/50. A heavier-treasury split (e.g., 70/30) raises the floor faster but starves buy-and-burn, suppressing the AMM spot dynamics that sustain secondary-market interest. A heavier-burn split (e.g., 30/70) pumps spot harder but compounds the floor more slowly. The 50/50 choice balances the two; we examine the trade-off as a sensitivity sweep in Section 6.4 (Figure 4 reports the resulting Pareto frontier) but do not claim it is Pareto-optimal in every utility specification. The simplicity of a constant ratio is itself a feature: it makes the router trivial to audit and removes a governance-attack surface that pre-existing tokenomic designs (Section 2.2) have repeatedly failed.

3.6 Permissionless fee collection

V4 does not store realized LP fees in a per-position slot; instead, every swap folds its input-side fee into a global accumulator that, for our purposes, accumulates the raw token-side and stable-side fee mass (Φ_t, Φ_s) until the position holder realizes it (Section 4.1 reconciles the raw-token framing with v4’s Q128.128 representation). To realize fees, a position holder must call `PoolManager.modifyLiquidity` with a zero liquidity delta, which returns a `BalanceDelta` settled through v4’s flash-accounting `take/settle` flow.

M² drives realization through a public, permissionless function:

```
function collectFees() external returns (uint256 stableOut, uint256 tokenOut);
```

The function acquires the pool-manager lock via `unlock`, calls `modifyLiquidity(..., liquidityDelta: 0, ...)` on the protocol-owned position, and obtains $(K, U) = (\Phi_t, \Phi_s)$, the realized token-side and stable-side fee amounts drawn from the raw accumulators. It then distributes the proceeds:

$$\text{caller bounty (token): } K_b = \lfloor 0.0025 \cdot K \rfloor, \quad (3)$$

$$\text{caller bounty (stable): } U_b = \lfloor 0.0025 \cdot U \rfloor, \quad (4)$$

$$\text{token burn: } K_{\text{burn}} = K - K_b, \quad (5)$$

$$\text{stable to treasury: } U_{\text{treas}} = U - U_b. \quad (6)$$

Critically, `collectFees` does *not* route through the revenue router: the router’s 50/50 split applies only to direct business-revenue inflows. The collected-fees path has its own simpler rule (stable to treasury, token to burn) because both destinations are unambiguously floor-positive regardless of market state.

Why permissionless with a caller bounty. The protocol cannot self-execute on a schedule without a keeper. A 0.25%-per-side bounty creates an economic incentive for any motivated party—MEV searchers, arbitrageurs, or holders themselves—to call the function whenever accumulated fees exceed the gas cost plus bounty threshold. The protocol thus self-drives. Theorem 9 (Section 5) formalizes the calling cadence as a Bertrand-style competitive race under atomless caller entry and derives the equilibrium inter-call interval Δt^* in closed form. The 0.25% rate is itself a design constant: too low and the bounty is dominated by gas cost in any

reasonable fee-volume regime; too high and meaningful value leaks to extractors. We do not optimize over this constant in the paper but report its sensitivity in the empirical follow-up.

3.7 Genesis ceremony and the floor-spot constraint

The genesis transaction performs five operations atomically: (i) deploy the four contracts; (ii) mint $S_0 = 10^9$ tokens to the deployer; (iii) deposit T_0 into the treasury; (iv) create the v4 pool with the dynamic-fee flag and add a single *full-range* position (over $[\text{MIN_TICK}, \text{MAX_TICK}]$) holding $L_{t,0} = 7.5 \times 10^8$ tokens (75% of supply) and $L_{s,0}$ stable; (v) transfer the remaining 2.5×10^8 tokens (25%) to a vesting vault. The deployer-chosen seeds T_0 and $L_{s,0}$ are constrained by

$$\frac{T_0}{S_0} = \frac{L_{s,0}}{L_{t,0}}, \quad (7)$$

which forces $F_0 = P_0$ at launch. The canonical instance studied in this paper uses $T_0 = \$1\text{M}$, $L_{s,0} = \$750\text{k}$, giving $F_0 = P_0 = \$0.001$ per token. The 75/25 supply allocation is fixed by the design; the absolute seed sizes are deployer-chosen subject to (7).

Why 75/25 rather than 50/50 or 90/10. A higher LP fraction extends the AMM’s effective lifetime under any fixed buy-and-burn cadence (Theorem 8: $n_{1/2} = L_{s,0}/B$ months under deterministic monthly buy size B). A lower LP fraction reserves more supply for the vesting vault, which supports team incentives and any post-launch airdrop. We treat the 75/25 split as an interior compromise and explore its sensitivity in Section 6.4, with the resulting LP-longevity-vs-floor-growth Pareto frontier in Figure 4.

3.8 Threat model

The formal guarantees of Section 4 hold under a specific set of assumptions about adversary capabilities, the backing stablecoin’s solvency, and the underlying execution layer. Table 2 delimits which adversaries are in scope, what they can do, and what is explicitly out of scope. We adopt the convention of Babel et al. [2023]: an attack vector is *in scope* if the protocol’s invariants are claimed to hold against it, and *out of scope* if violation under that vector is acknowledged as residual risk requiring a separate analysis.

Table 2: Threat model. “In scope” means the protocol’s invariants are claimed to hold against the indicated capability; “out of scope” acknowledges residual risk requiring a separate analysis. The table delimits the boundary of the formal guarantees of Section 4.

Adversary	Capability	Scope	Where treated
Rational holder	Choose any composition of redeem, LP swap, transfer; observe public state	In scope	Thm 2, Thm 4
Coordinated coalition (any size up to total supply)	Submit any sequence of in-protocol operations; coordinate timing	In scope (simultaneous-strategy class)	Thm 6; sequenced-class supremum is Section 8.5 (P7)
MEV searcher (single block)	Reorder, sandwich, frontrun any user transaction within a block; pay priority gas	In scope (bounded leakage)	Thm 10; §3.4 mitigations
collectFees-adjacent MEV (single block)	Time collectFees relative to large swaps within a single block to capture floor-step value	In scope (single-block)	Cor. 3 (closure under re-ordering); §3.4

continued on next page

Table 2 continued from previous page.

Adversary	Capability	Scope	Where treated
collectFees-adjacent MEV (cross-block)	Sequence collectFees across blocks to interact with attacker redemptions	Out of scope	§8 (P4); special case of multi-block MEV
MEV searcher (multi-block, builder collusion)	Coordinate across blocks; influence builder choice	Out of scope	§8 (P2, P4)
Operator (revenue depositor)	Choose timing and magnitude of routeRevenue; choose not to call	Partially in scope: floor is preserved at zero revenue; growth is operator-dependent	§5 (CEF analog under cessation); §8.3
Operator (strategic timing with vested holdings)	Time routeRevenue deposits jointly with private vested-token positions (25% of supply); exploit information advantage from knowing imminent deposits	Partially in scope: within-block compositions bounded by Thm 6 (simultaneous-strategy class); cross-block strategic timing is treated under (P7), and the operator IC analysis is open	§8.5 (P7); §8.3 (operator IC)
Operator (post-genesis bytecode change)	N/A—no upgrade path exists	N/A	Immutability is structural
Vesting recipient	Dump vested tokens to LP after cliff	In scope	Symmetric 3% sell fee applies; Thm 6 applies
Third-party oracle consumer	External protocol uses M ² ’s spot as a price input and is exposed to spot manipulation by an adversary	Out of scope (the on-chain protocol cannot prevent third-party misuse)	§8 (P3); deployer-side documentation
Backing-stable issuer	Freeze treasury balance; depeg the stable; halt transfers	Out of scope (exogenous)	§8; USDC March-2023 depeg as canonical event
Smart-contract bug (any contract in the system)	Violate any stated invariant via implementation defect	Out of scope (no mechanism for post-deployment fix)	§8; audit + mechanized-verification path
L1/L2 consensus failure	Halt the chain; censor transactions long-term	Out of scope (substrate assumption)	§8

What the threat model does not promise. M² is *trust-minimized*, not trustless. The rows of Table 2 marked “out of scope” enumerate the residual surfaces. In particular, we make no attempt to defend against an issuer of the backing stable who freezes the treasury’s balance; against a smart-contract bug that we and our auditors fail to find; against multi-block MEV

under builder collusion (a class of attacks for which no current AMM design has defense); against operator fraud at the revenue-deposit layer (the operator can always choose not to deposit, in which case the floor is preserved but does not grow); against cross-block `collectFees`-adjacent MEV; or against third-party oracle consumers who treat M^2 's spot as a price input (the on-chain protocol cannot prevent third-party misuse). The contributions of this paper concern the operations the protocol *does* define; everything outside that surface is residual risk acknowledged in Section 8.

4 Formal Invariants and Proofs

This section states the floor-monotonicity property formally, proves it by case analysis over the protocol's complete operation set, and establishes the integer-arithmetic redemption lemma on which the cleanest version of the redemption case relies. We also state the redemption-fairness corollary, a directional conjecture inverting the standard AMM-LP loss-versus-rebalancing framing of [Milionis et al. \[2024\]](#) (Conjecture 5), and a closure result under arbitrary operation interleaving. We close with a sketch of the mechanized-verification path we are pursuing.

4.1 State model and the operation set

We model the system state as the tuple

$$\sigma = (T, S, L_t, L_s, \Phi_t, \Phi_s), \quad (8)$$

where $T \in \mathbb{Z}_{\geq 0}$ is the treasury balance in the stable's smallest unit ($Q_s = 10^{d_s}$ units per dollar); $S \in \mathbb{Z}_{\geq 0}$ is the total token supply in 10^{-18} -token units; $(L_t, L_s) \in \mathbb{Z}_{\geq 0}^2$ are the LP's active reserves in the same units; and $(\Phi_t, \Phi_s) \in \mathbb{Z}_{\geq 0}^2$ are the *raw unrealized fee mass* on the protocol's LP position—i.e. the token-side and stable-side fee amounts awaiting realization to the protocol's books on the next `collectFees` call, denominated in the same units as S and T respectively.² On a swap of token input N_{in} with sell fee f_s , the input-side fee $f_s N_{\text{in}}$ is added to Φ_t ; on a swap of stable input X_{in} with buy fee f_b , the input-side fee $f_b X_{\text{in}}$ is added to Φ_s . At a `collectFees` call the accumulated (Φ_t, Φ_s) are drawn off as the realized fees $(K, U) = (\Phi_t, \Phi_s)$ and both accumulators are reset to zero.

The floor and spot prices are derived quantities, $F = T/S$ (a rational number, evaluated faithfully to integer truncation by the on-chain expression (1)) and $P = L_s/L_t$. The constant-product invariant $k = L_t \cdot L_s$ is preserved by every swap up to the asymmetric-fee deduction described in Section 3.4; this identity is exact (not an approximation), because the protocol-owned v4 position is full-range, so (L_t, L_s) are the V2-style reserves of a single piecewise-constant liquidity segment with no tick crossings (Section 3.4, “Full-range position and V2-equivalence”).

The protocol's *operation set* Ops is the disjoint union of seven operation classes, exhausting every state transition any externally-callable entry point can induce:

$$\begin{aligned} \text{Ops} = \{ & \text{RevToTreasury}(X), \text{BuyAndBurn}(X), \text{Redeem}(N), \\ & \text{LPBuy}(X_{\text{in}}), \text{LPSell}(N_{\text{in}}), \text{Transfer}(N, a \rightarrow b), \text{CollectFees}() \}. \end{aligned} \quad (9)$$

The first two are the two halves of `routeRevenue`; we split them because they affect the state differently. The next three are the holder-facing operations; the sixth is an ERC-20 transfer, listed for completeness; the seventh realizes accrued fees. No other entry point exists in the protocol bytecode. The threat model of Section 3.8 confines adversary capabilities to compositions of elements of Ops with arbitrary timing and ordering.

²The Uniswap v4 implementation does not store Φ_t, Φ_s as raw token amounts; instead, it maintains per-currency Q128.128 fee-growth accumulators `feeGrowthGlobalX128` into which each swap's input-side fee is folded, with the position's claim computed as $K_{\text{real}} = (\text{accumulator delta}) \cdot \text{liquidity} / 2^{128}$ at realization. Throughout this paper we use the raw-token framing (Φ_t, Φ_s) for dimensional clarity of the proofs; the v4 Q128.128 realization is an implementation detail that the algebra abstracts away.

4.2 Integer-arithmetic redemption

The cleanest version of the redemption case requires a small lemma about how the integer arithmetic in the contract implementation relates to the rational ideal $F = T/S$. We state it for a redemption of N tokens against state (T, S) with $N \leq S$.

Lemma 1 (Integer-Arithmetic Redemption). *Let $(T, S) \in \mathbb{Z}_{>0}^2$ with $N \in \mathbb{Z}_{>0}$ and $N \leq S$. Let the on-chain redemption payout be the integer*

$$P = \left\lfloor \frac{N \cdot T}{S} \right\rfloor, \quad \text{and write } r = (N \cdot T) \bmod S \in \{0, 1, \dots, S-1\}. \quad (10)$$

Let $T' = T - P$ and $S' = S - N$. Then either $S' = 0$ —in which case $N = S$ forces $r = 0$ and $P = T$, so $T' = 0$ exactly—or $S' > 0$ and

$$\frac{T'}{S'} = \frac{T}{S} + \frac{r}{S \cdot S'} \geq \frac{T}{S}. \quad (11)$$

In particular, $F_{\text{new}} \geq F_{\text{old}}$, with equality if and only if $S \mid N \cdot T$.

Proof. By definition of integer division, $N \cdot T = S \cdot P + r$ with $0 \leq r < S$. Then

$$T - P = T - \frac{N \cdot T - r}{S} = \frac{T(S - N) + r}{S} = \frac{T \cdot S' + r}{S}.$$

If $S' > 0$, dividing by S' gives $T'/S' = T/S + r/(S \cdot S')$, which is (11). The right-hand correction term is non-negative; it is zero exactly when $r = 0$, i.e. when S divides $N \cdot T$ exactly. If $S' = 0$, then $N = S$ and $N \cdot T = S \cdot T$ is divisible by S , so $r = 0$ and $P = T$; hence $T' = T - P = 0$ exactly. This is the terminal redemption that drains the contract; the floor at this point is undefined by Section 3.2, and the contract reverts on the next `floorPrice()` call via the `SupplyExhausted` error. $\square$

The lemma sharpens the informal “invariant up to truncation” reading of Case 3 to an exact inequality: integer truncation in the redemption payout strictly favors the treasury and so can only raise the floor for remaining holders. The residual $r/(S \cdot S')$ is at most $1/S'$ per redemption call (a sub-unit-of-stable effect), but it is structurally one-directional.

4.3 Floor monotonicity

Theorem 2 (Floor Monotonicity). *Let σ be any reachable state of the protocol and let $op \in \text{Ops}$ be any operation applicable in σ . Let $\sigma' = op(\sigma)$. Then*

$$F(\sigma') \geq F(\sigma), \quad (12)$$

with the equality and strict-inequality cases as enumerated in the proof.

Proof. We argue by cases over the seven operation classes of (9). In each case we write $(T, S) \rightarrow (T', S')$ for the treasury and supply transition and compute the resulting floor.

Case 1 — RevToTreasury(X) with $X \geq 0$. The router moves X stable to the treasury. State transition: $(T + X, S)$. Floor: $(T + X)/S \geq T/S$, with strict inequality iff $X > 0$.

Case 2 — BuyAndBurn(X) with $X \geq 0$. The router uses X stable to execute a buy on the LP, receiving and burning $Y(X) = L_t - k/(L_s + (1 - f_b)X)$ tokens, where $f_b = 0.001$ is the buy-side hook fee. State transition: $(T, S - Y(X))$. Floor: $T/(S - Y(X)) \geq T/S$, with strict inequality iff $Y(X) > 0$, which holds for any $X > 0$ provided L_s is finite and positive.

Case 3 — Redeem(N) with $1 \leq N \leq S$. By Lemma 1, $F_{\text{new}} \geq F_{\text{old}}$, with equality iff $S \mid N \cdot T$. For the abstract rational ideal $F = T/S$, the floor is preserved exactly; under integer truncation, the floor strictly rises by a sub-unit residual that accrues to the treasury.

Case 4 — LPBuy(X_{in}) from an external user. The user sends X_{in} stable to the LP and receives tokens out. The LP reserves change; the treasury and total supply are untouched. State transition: (T, S) , so $F' = F$. Equality holds in every instance.

Case 5 — LPSell(N_{in}) from an external user, with the hook's 3% fee on input. Let $f_s = 0.03$ be the sell-side fee. The hook deducts $f_s \cdot N_{\text{in}}$ from the input before the curve, folding the deducted amount into Φ_t . The remaining $(1 - f_s)N_{\text{in}}$ swaps through the curve. State transition: $(T, S, L_t + (1 - f_s)N_{\text{in}}, L_s - \Delta L_s, \Phi_t + f_s N_{\text{in}}, \Phi_s)$, where $\Delta L_s = L_s - k / (L_t + (1 - f_s)N_{\text{in}})$ is the stable amount sent to the seller. The treasury and total supply are unchanged, so $F' = F$ at the moment of the swap. The fee tokens reside in the v4 fee accumulator and are not yet on the protocol's books; they are realized in Case 7.

Case 6 — Transfer($N, a \rightarrow b$). An ERC-20 balance transfer between addresses a and b . Treasury, supply, and LP reserves are untouched. $F' = F$ trivially.

Case 7 — CollectFees(). The function realizes the accrued unrealized fee mass as $(K, U) = (\Phi_t, \Phi_s)$ and resets both accumulators to zero (the v4 implementation extracts the same quantities from the Q128.128 accumulator scaled by the position's liquidity; see footnote in Section 4.1). The token-side K is split into $K_b = \lfloor 0.0025K \rfloor$ (caller bounty, remains in circulation) and $K_{\text{burn}} = K - K_b$ (burned). The stable-side U is split into $U_b = \lfloor 0.0025U \rfloor$ (caller bounty) and $U_{\text{treas}} = U - U_b$ (added to treasury). State transition: $(T + U_{\text{treas}}, S - K_{\text{burn}}, L_t, L_s, 0, 0)$. New floor:

$$F' = \frac{T + U_{\text{treas}}}{S - K_{\text{burn}}} \geq \frac{T}{S} = F,$$

with strict inequality iff $U_{\text{treas}} > 0$ or $K_{\text{burn}} > 0$, which holds whenever any swap has occurred since the previous `collectFees` call.

Exhaustiveness. The seven cases exhaust Ops as defined in (9). The protocol bytecode exposes no other externally callable function that modifies (T, S, L_t, L_s) : the token has `{redeem, burn (restricted), transfer, transferFrom, approve, permit}`; the treasury has `{payRedemption (restricted)}`; the router has `{routeRevenue (restricted)}`; the hook has `{beforeSwap, collectFees}`. Approval and permit do not modify the state tuple σ ; `beforeSwap` runs as part of a swap (subsumed by Cases 4 and 5); the restricted functions can be invoked only by the listed callers and so are subsumed by the protocol-initiated operations enumerated above. No other path can reach σ .³

Combining all cases: $F(\sigma') \geq F(\sigma)$ holds for every $\text{op} \in \text{Ops}$, with strict inequality in Cases 1, 2, 7 (and weakly in Case 3 under the abstract ideal, strictly under integer arithmetic with $r > 0$), and equality in Cases 4–6. $\square$

4.4 Closure under composition

Corollary 3 (Closure under Arbitrary Interleaving). *For any finite sequence $\text{op}_1, \text{op}_2, \dots, \text{op}_m \in \text{Ops}$ applied to a reachable state σ , the resulting state $\sigma_m = \text{op}_m \circ \dots \circ \text{op}_1(\sigma)$ satisfies $F(\sigma_m) \geq F(\sigma)$.*

Proof. By induction on m . The base case $m = 0$ is trivial. For the inductive step, $F(\sigma_m) \geq F(\sigma_{m-1}) \geq F(\sigma)$ by Theorem 2 and the induction hypothesis. $\square$

Corollary 3 matters because it absorbs the intra-block ordering ambiguity exploited by MEV searchers. Any reordering or insertion of in-protocol operations between Case 5's swap and Case 7's realization can only weakly raise the floor at the realization moment; no adversarial schedule of Ops elements can produce $F(\sigma_m) < F(\sigma)$. This is distinct from the standard MEV-victim framing in which sandwich attacks degrade a target user's outcome—in M^2 , the

³The hook's `initializePool` is a one-shot genesis call that transitions (L_t, L_s) from zero to the genesis seed (Section 3.7) and reverts on every subsequent call; `unlockCallback` is `onlyPoolManager`-gated and its σ -mutations are subsumed by Cases 2, 4, 5, and 7 (router buy-and-burn and AMM swaps re-enter the hook through this callback). Neither path introduces a state transition outside Ops.

floor invariant is robust to all such reorderings; what an MEV searcher can extract is a portion of the swap surplus that would otherwise have accrued to the LP, not value from the floor.

4.5 Redemption fairness

Theorem 4 (Redemption Fairness, No First-Mover Advantage). *Let two holders, a and b , simultaneously submit redemption transactions for N_a, N_b tokens against a state σ . Let any miner-/builder-chosen ordering execute these (and possibly other in-protocol operations) sequentially, yielding payouts $P_a^{(\pi)}, P_b^{(\pi)}$ under ordering π . Then for every ordering π in which a executes before b :*

$$P_a^{(\pi)} / N_a \leq P_b^{(\pi)} / N_b. \quad (13)$$

That is, the per-token payout for the later redeemer is weakly higher than that of the earlier redeemer; no first-mover advantage exists in floor terms.

Proof. The per-token payout for a redemption is the floor evaluated at that redemption’s block-state. By Theorem 2 Case 3 (and Lemma 1 for the integer-arithmetic refinement), a ’s redemption preserves or raises the floor; by Corollary 3, any intervening operations preserve or raise it further. Therefore b ’s redemption evaluates the floor at a value at least as high as a ’s did, establishing (13). $\square$

Contrast with classical bank-run dynamics. The Chen et al. [2010] first-mover advantage is structurally absent in M^2 ; the formal contrast appears in Section 5.2.

What the theorem does not promise. Theorem 4 concerns the floor, not the absolute dollar amount a holder receives in alternative dollar terms (e.g., spot-market value of the burned tokens). A holder who could have sold at spot above the floor accepts an opportunity cost by redeeming. The theorem also does not preclude an MEV-style reordering in which a ’s submitted redemption is preceded by an opportunistically-inserted operation; under Corollary 3, such an insertion can only *help* the redeemer relative to their own submission-state evaluation, because the floor can only weakly rise.

4.6 LVR / floor-capture identity

The Loss-Versus-Rebalancing (LVR) framework of Milionis et al. [2024] characterizes the cost an AMM LP incurs from supplying liquidity against informed arbitrageurs trading on an external venue. In the standard framing, LVR is a loss to the LP that is captured by arbitrageurs. For the protocol-owned LP position in M^2 , this framing appears to invert. We state the inversion as a *conjecture* rather than a theorem: the proof of a quantitative identity requires a careful unit reconciliation between LVR’s per-unit-time dollar-flow formulation and M^2 ’s cumulative-floor formulation that we have not yet completed.

Conjecture 5 (LVR / Floor-Capture Identity). *Let $\text{LVR}_{[t_0, t_1]}$ denote the cumulative loss-versus-rebalancing of the protocol-owned LP position over an interval $[t_0, t_1]$, in the sense of Milionis et al. [2024]. Under the asymmetric-fee mechanism of Section 3.4 with $f_b = 0.001$ on buys and $f_s = 0.03$ on sells, the LVR-equivalent value of the protocol-owned LP position is recovered as floor growth via the Case 5 plus Case 7 composition of Theorem 2, modulo a leakage of at most 0.25% per side at `collectFees` realization. In particular, on any interval $[t_0, t_1]$ over which `collectFees` is invoked at least once,*

$$\Delta F_{[t_0, t_1]} \cdot S(t_1) \gtrsim (1 - 0.0025) \cdot \text{LVR}_{[t_0, t_1]}^{\text{fee-attributable}}, \quad (14)$$

where $\Delta F_{[t_0, t_1]}$ is the cumulative floor increase attributable to swap-induced fee growth realized via `collectFees` in the same interval, and the right-hand side is the portion of cumulative LVR that the asymmetric fee mechanism converts into protocol-realizable fee growth.

Discussion. The conjecture is directional rather than a quantitative identity. The intuition is the standard inversion: in the canonical LP-vs-arbitrageur setting LVR is value extracted from the LP by an external trader; for the protocol-owned LP the same swap surplus is folded into the v4 fee accumulator and realized to the treasury (stable side) or burned (token side) at the next `collectFees`, raising $F = T/S$ on both legs. Converting (14) into a theorem requires three steps we have not completed: (i) a unit reconciliation between the continuous-time stochastic-calculus framing of LVR (an instantaneous dollar flow per unit of LP liquidity) and the discrete event-driven framing of F growth at `collectFees` ticks; (ii) a careful accounting of the per-swap fee asymmetry, which makes the buy-leg and sell-leg contributions to ΔF structurally different (the buy fee accrues to Φ_s and routes to treasury; the sell fee accrues to Φ_t and routes to burn); and (iii) verifying that no value leaks outside the (fee-collection bounty, MEV) channels already accounted for in Theorems 2 and 10. We identify the completion of this proof as an open problem (Section 8.5, problem (P6)) and note that even the directional conjecture is sufficient for the qualitative argument in Section 2.3: the protocol-owned LP is the structural object that converts LVR from a loss-to-LP into floor growth captured by the holder base.

4.7 Mechanized verification path

We have given a paper proof. End-to-end mechanized verification of Theorem 2 against the deployed Solidity bytecode is a separate exercise, which we sketch here and defer to follow-up work.

Three tool stacks are candidates. **Halmos** [Tolmach et al., 2022] performs symbolic execution of EVM bytecode and discharges quantified properties of the form “for all reachable states, P holds.” The floor invariant fits this template: we specify the property as a postcondition over (T, S) and run Halmos on each entry point. **Certora’s Prover** expresses invariants in CVL and proves them by SMT-backed symbolic execution; its strength is composing per-function rules into contract-level invariants via the `invariant` keyword. **The K framework** [Tolmach et al., 2022] gives the deepest semantics-level guarantee at the cost of greater specification overhead.

The non-trivial cases for any of these tools are Cases 5 and 7, where the floor invariant holds only after the temporal pairing of the swap with the subsequent `collectFees`. We propose specifying this as a two-state invariant over reachable pairs $(\sigma_t, \sigma_{t+\text{collect}})$, with the intermediate-state property weakened to “floor non-decreasing modulo accrued fee growth.” This is a known idiom in the formal-verification literature and is supported by all three stacks. We will report mechanized results in the empirical follow-up paper.

5 Economic Analysis

The proofs of Section 4 establish a structural floor-monotonicity property: no protocol-defined operation can lower the floor. They do not yet address *what an adversary chooses to do* when faced with the protocol. This section formalizes the adversarial decision problem as a redemption-versus-dump coordination game (a structural counterpoint to Diamond–Dybvig, whose multiple-equilibrium dynamics fail to obtain in M^2), characterizes the profit-maximizing composition of LP-dump and redemption strategies, derives the closed-form saturation point above which the dump-portion of any optimal strategy is bounded, contrasts the resulting dynamics with the Terra–Luna collapse [Liu et al., 2023], and analyzes the system’s behavior under operator-revenue cessation through a closed-end-fund analog [Cherkes et al., 2009]. We also state the LP-exhaustion, MEV-leakage, and collect-fees-equilibrium results referenced in Section 3.

5.1 The two exit paths

A holder of N tokens, at state $\sigma = (T, S, L_t, L_s)$, has two protocol-defined exit channels. The *redemption channel* burns the N tokens and pays out $N \cdot F$ stable from the treasury; by Theorem 2 Case 3, this preserves the floor exactly (and strictly raises it by a sub-unit residual under integer arithmetic; Lemma 1). The *LP-dump channel* routes the N tokens through the AMM's sell side, where the hook deducts a $f_s = 3\%$ fee on input before the constant-product curve is applied. The holder may also mix the two strategies—redeem a fraction $1 - \alpha$ and dump a fraction α , for any $\alpha \in [0, 1]$. The adversary's decision problem is to choose α to maximize total stable proceeds. We restrict attention throughout this subsection to *simultaneous* strategies in which the attacker's dump and redemption occur in a single block, without an intervening `collectFees` realization; the strictly more permissive sequenced class is treated in the discussion following Theorem 6.

Writing $M = \alpha N$ for the tokens routed to the LP, the LP payout function is

$$\pi_{\text{LP}}(M) = L_s - \frac{k}{L_t + (1 - f_s)M}, \quad (15)$$

which is strictly concave in M on $M \geq 0$, with $\pi_{\text{LP}}(0) = 0$, $\pi'_{\text{LP}}(0) = (1 - f_s)L_s/L_t = (1 - f_s)P$, and $\lim_{M \rightarrow \infty} \pi_{\text{LP}}(M) = L_s$. The redemption payout is linear: $\pi_{\text{R}}(N - M) = (N - M)F$. The total attacker yield as a function of the strategy α is

$$\Pi(\alpha; N, \sigma) = \pi_{\text{LP}}(\alpha N) + (1 - \alpha)NF. \quad (16)$$

Π is concave in α on $[0, 1]$ (the LP term is concave; the redemption term is linear), so the maximum is attained either at a boundary or at the unique interior critical point.

5.2 Saturation and the run-resistance theorem

Differentiating (16) in α and setting the result to zero gives the interior optimum,

$$(L_t + (1 - f_s)\alpha^* N)^2 = \frac{k(1 - f_s)}{F}, \quad (17)$$

which after rearrangement yields the saturation quantity

$$A^* := \alpha^* N = \frac{1}{1 - f_s} \left(\sqrt{\frac{k(1 - f_s)}{F}} - L_t \right). \quad (18)$$

The key structural fact is that A^* depends only on the state σ , *not on the adversary's holdings* N . The optimal dump quantity is a property of the AMM and the floor, not of the attacker's size.

Theorem 6 (Composite-Strategy Bank Run with Saturation). *Within the simultaneous-strategy class of Section 5.1, at state σ , let A^* be the saturation quantity (18) and let $\Pi^*(N; \sigma) = \max_{\alpha \in [0, 1]} \Pi(\alpha; N, \sigma)$ be the attacker's value function on holdings N . Then:*

0. (Pre-saturation regime.) *If $A^* \leq 0$ —equivalently $(1 - f_s)P \leq F$, a near-launch or fee-suppressed-spot state in which the post-fee LP spot does not exceed the floor—the optimum is $\alpha^* = 0$ (pure redemption); pure redemption strictly dominates any positive dump under the asymmetric sell fee. The value is $\Pi^*(N) = NF$.*
1. (Pure-dump regime.) *If $0 < A^*$ and $N \leq A^*$, the optimum is $\alpha^* = 1$ (pure LP dump). The value is $\Pi^*(N) = \pi_{\text{LP}}(N) = L_s - k/(L_t + (1 - f_s)N)$.*

2. (Mixed regime.) If $N > A^*$, the optimum is $\alpha^* = A^*/N \in (0, 1)$, with dump quantity exactly A^* and redemption quantity $N - A^*$. The value is

$$\Pi^*(N) = (N - A^*)F + \left[L_s - \sqrt{kF/(1 - f_s)} \right]. \quad (19)$$

3. (Bounded run premium.) For any $N > A^*$, the run premium over pure redemption,

$$\Delta^* := \Pi^*(N) - NF = \left[L_s - \sqrt{kF/(1 - f_s)} \right] - A^*F, \quad (20)$$

is a state-only constant, independent of N . The expression admits the equivalent closed forms

$$\Delta^* = \left(\sqrt{L_s} - \sqrt{L_t F / (1 - f_s)} \right)^2 = L_s \cdot \left(1 - \sqrt{F / ((1 - f_s)P)} \right)^2, \quad (21)$$

which expose two structural facts: Δ^* vanishes whenever the post-fee LP spot $(1 - f_s)P$ approaches the floor F , and the bound is structurally tight rather than $\Delta^* \leq L_s$ in the worst case. At the canonical month-12 state, $F / ((1 - f_s)P) \approx 0.7637$ and $\Delta^*/L_s \approx 1.59\%$ (not 100%).

4. (Asymptotic redemption.) $\Pi^*(N)/N \rightarrow F$ as $N \rightarrow \infty$. The marginal value of an additional held token, beyond what pure redemption returns, decays to zero.

Proof. By concavity of Π in α , the unconstrained maximizer (17) is the unique critical point. If $\alpha^* \geq 1$ (equivalently $A^* \geq N$), the constraint binds at $\alpha = 1$, giving regime 1. Otherwise the interior solution gives regime 2 with dump quantity $\alpha^*N = A^*$, a state-only constant. Substituting $\alpha^* = A^*/N$ into (16) and simplifying using $L_t + (1 - f_s)A^* = \sqrt{k(1 - f_s)/F}$ from (17) yields (19). The run premium (20) is the constant tail in (19) after subtracting NF . To obtain the closed form (21), set $\beta = \sqrt{(1 - f_s)/F}$ and use $k = L_t L_s$ to write $\sqrt{kF/(1 - f_s)} = \sqrt{L_t L_s}/\beta$ and $A^*F = (1/\beta^2)(\beta\sqrt{L_t L_s} - L_t) = \sqrt{L_t L_s}/\beta - L_t/\beta^2$, so

$$\Delta^* = L_s - \frac{2\sqrt{L_t L_s}}{\beta} + \frac{L_t}{\beta^2} = \left(\sqrt{L_s} - \sqrt{L_t}/\beta \right)^2 = \left(\sqrt{L_s} - \sqrt{L_t F / (1 - f_s)} \right)^2.$$

The second form in (21) follows by factoring $\sqrt{L_s}$ and using $P = L_s/L_t$. The asymptote in (4) is immediate from $\Pi^*(N)/N = F + \Delta^*/N \rightarrow F$. $\square$

Structural interpretation of Δ^* . The closed form (21) shows that Δ^* is the LP’s “excess depth above the no-arbitrage curve” at the asymmetric-fee-adjusted spot price. It scales with how far the post-fee spot $(1 - f_s)P$ exceeds the floor F , not with L_s in the worst case: when the spot has been compressed by sustained selling so that $(1 - f_s)P \rightarrow F$, the structural advantage of dumping over pure redemption vanishes entirely.

Sequenced strategies. The simultaneous-strategy restriction of Theorem 6 excludes strategies in which the attacker interleaves `collectFees` between the dump and the redemption. In a sequenced strategy, the attacker dumps A^* tokens, waits (or pays) for a `collectFees` call that realizes the token-side fees the attacker’s own dump just deposited into Φ_t , and only then redeems the remaining $N - A^*$ tokens against a now-higher floor. The marginal benefit over the simultaneous strategy is the floor increment ΔF realized between dump and redemption, multiplied by the attacker’s residual holdings $N - A^*$. At the canonical month-12 state, the dump of $A^* \approx 6.20 \cdot 10^7$ tokens deposits $f_s A^* \approx 1.860 \cdot 10^6$ tokens into Φ_t , of which a fraction $1 - 0.0025$ (the caller bounty leaks the rest) is burned at the next `collectFees`; supply falls by $\approx 1.855 \cdot 10^6$ tokens, and the floor rises by $\Delta F = T/(S - 1.855 \cdot 10^6) - T/S \approx \$6.698 \cdot 10^{-6}$ per token. The maximal sequenced premium, attained when the attacker initially holds the entire

circulating supply $N = S$, is $(N - A^*) \cdot \Delta F \approx 6.047 \cdot 10^8 \cdot \$6.698 \cdot 10^{-6} \approx \$4,050$, an 18.9% improvement on Δ^* . Two structural observations: (i) the sequenced benefit *is* a function of the attacker’s holdings N , so the “state-only constant” claim of Theorem 6(3) does not extend to the sequenced class; (ii) the per-held-token benefit is bounded above by ΔF (a state-only quantity), so the simultaneous-class asymptote $\Pi^*(N)/N \rightarrow F$ of part (4) is replaced in the sequenced class by $\Pi_{\text{seq}}^*(N)/N \rightarrow F + \Delta F$ —a strictly higher but still state-only-bounded asymptote. At canonical state this is $F + 6.7 \cdot 10^{-6}$, a 0.28% asymptotic premium per token. The qualitative “no first-mover advantage” content of Theorem 4 is preserved, since the bound on the per-token gain remains independent of N . The closed-form supremum across the full sequenced strategy class—in which the attacker may interleave LP dumps, `collectFees` calls, and redemptions within or across blocks—is left as an open problem (Section 8.5, P7); a full treatment requires modeling the equilibrium response of the `collectFees`-bounty payer and the v4 fee-realization timing dynamics.

Numerical illustration at the canonical month-12 state. At the values produced by the deterministic 12-month projection of Section 6— $T = \$1.6\text{M}$, $S = 2 \cdot 10^9/3 \approx 6.667 \cdot 10^8$, $F = \$0.0024$ exactly, $L_t = 1.25 \cdot 10^9/3 \approx 4.167 \cdot 10^8$, $L_s = \$1.35\text{M}$, $k = L_t L_s = 5.625 \cdot 10^{14}$, $f_s = 0.03$ —equation (18) gives

$$\sqrt{\frac{k(1-f_s)}{F}} = \sqrt{2.2734 \cdot 10^{17}} \approx 4.76806 \cdot 10^8, \quad A^* \approx 6.1999 \cdot 10^7 \text{ tokens},$$

which is about 9.30% of the circulating supply. Equation (20) then evaluates as

$$\Delta^* = \underbrace{\$1,350,000}_{L_s} - \underbrace{\sqrt{\frac{kF}{1-f_s}}}_{\approx \$1,179,725.64} - \underbrace{A^* F}_{\approx \$148,797.80} \approx \$21,476.56.$$

The closed-form value, computed in 60-digit decimal arithmetic, is $\Delta^* = \$21,476.5621\dots$; the figures in the paper are rounded to the nearest cent. Even an adversary with the entire circulating supply $N = S$ can extract at most $NF + \Delta^* \approx \$1,621,476.56$ from the system—this is the treasury balance plus a residual constant bounded above by L_s .

Pure-strategy crossover. A naive pure-strategy analysis would equate the two pure-strategy yields $\pi_{\text{LP}}(N) = NF$ to find a crossover N^{cross} where pure dump and pure redemption yield the same proceeds; numerically at the canonical month-12 state, $N^{\text{cross}} \approx 1.33 \cdot 10^8$. Theorem 6 subsumes this: the saturated mixed strategy of regime (2) yields more than either pure strategy whenever $N > A^*$, so the pure-strategy crossover is informationally inferior to (20) as a characterization of the attacker’s optimal yield.

Diamond–Dybvig contrast. The classical [Diamond and Dybvig \[1983\]](#) model produces multiple-equilibrium bank-run dynamics because the bank’s reserves are finite and patient depositors strictly lose if impatient depositors redeem first. In M^2 , by Theorem 4, later redeemers weakly dominate earlier redeemers in floor terms; by Theorem 6(3), the total adversarial premium over redemption is bounded by a state-only constant; and by Theorem 2, no operation in Ops reduces the floor. The strategic-complementarity channel of [Chen et al. \[2010\]](#) (early redeemers extract value from later ones) is structurally absent. The M^2 “run” is a single-equilibrium game in which the rational strategy converges to pure redemption as N grows, and the system’s terminal state under a complete-redemption attack is the same as its terminal state under no attack at all: the LP retains its k , the treasury drains in lockstep with supply burn, and the per-token floor is preserved.

5.3 Disanalogy with the Terra–Luna collapse

The Terra–Luna collapse [Liu et al., 2023, Briola et al., 2023] is the standard cautionary tale of an algorithmic-stablecoin system that entered a reflexive death spiral under coordinated redemption pressure. The mechanism: UST redemption minted LUNA at the prevailing LUNA price, so a coordinated UST sell increased LUNA supply, lowering LUNA price, which further raised the LUNA quantity minted per UST burned, and so on. The endogenous-collateral structure produced a positive feedback loop in which redemption amplified its own pressure.

The M^2 disanalogy is structural along three axes:

1. *Collateral exogeneity.* The M^2 treasury holds a single deployer-chosen USD-pegged ERC-20 (e.g. USDC, DAI). The protocol does not mint the backing stable. There is no endogenous-collateral feedback channel of the Terra–LUNA kind because the backing asset’s supply and price are not protocol-determined.
2. *One-way treasury inflow.* Redemption removes stable from the treasury only in lockstep with burning the redeemed tokens. The per-token claim $F = T/S$ is therefore invariant across redemption events, by Lemma 1. In Terra–LUNA, redemption changed both legs (UST supply down, LUNA supply up) by ratios that did not preserve any analogous invariant.
3. *No pegging mechanism.* M^2 is not a stablecoin and has no peg target. The AMM spot price floats; only the floor is invariant-protected. Terra–LUNA’s collapse depended on the loss of confidence in a peg target; M^2 has no target to lose.

The closest TradFi analog of M^2 is a closed-end fund holding U.S. dollars in custody with continuous NAV redemption [Lee et al., 1991, Cherkes et al., 2009]. The closest *negative* analog is Terra–LUNA. The crucial design choice that places M^2 in the first category and not the second is the exogeneity of the backing asset.

5.4 Operator-revenue cessation and the CEF analog

The growth of the floor depends on the operator’s continued routing of business revenue. If the operator stops depositing (whether by choice, bankruptcy, or third-party disruption), revenue inflow becomes zero. We characterize the system’s degradation under this regime through a closed-end-fund analog.

Theorem 7 (Spot–Floor Convergence under Zero Revenue). *Suppose the operator deposits no revenue from time t_0 onward. Let $D(t) := P(t) - F(t)$. Then:*

1. (Floor stickiness.) *$F(t)$ is non-decreasing in t (Theorem 2). Strict increases occur only on `collectFees` invocations realizing organic-flow fees.*
2. (Spot floor via arbitrage.) *Under any positive-probability arbitrageur path—arbitrageurs are available to buy on the LP whenever $P < F$ —we have $P(t) \geq F(t)$ for all $t \geq t_0$. Equivalently, $D(t) \geq 0$.*
3. (Convergence under net-sell pressure.) *Suppose that after t_0 the cumulative non-arbitrage sell volume strictly exceeds the cumulative non-arbitrage buy volume in expectation, while arbitrageurs remain available to buy whenever $P < F$ as in (2). Then $P(t)$ is non-increasing in expectation outside arbitrage events (non-arbitrage sells push it down; arbitrage entry prevents it from settling below the floor), $F(t)$ is non-decreasing, and $D(t)$ converges to a limit bounded above by the arbitrage threshold $f_b \cdot F \approx 10$ basis points of F (the minimum gap at which a one-dollar-sized arbitrage round-trip is profitable; smaller trades face proportionally larger relative gas costs).*

Proof sketch. (1) Direct from Theorem 2: in the zero-revenue regime, only Cases 3, 4, 5, 6, 7 of the operation set are exercised, and none lowers the floor. (2) Suppose for contradiction $P(t) < F(t)$. An arbitrageur can buy on the LP at P (paying the 0.10% buy fee), redeem against the treasury at F , and net a profit of $(F - P)/P - 0.001$ per dollar deployed—positive whenever the gap exceeds 10 basis points. The buy raises P and the redemption preserves F ; the gap is closed by the assumed positive-probability path. (3) Non-arbitrage sells lower P along the AMM curve; the floor either stays flat or rises on `collectFees`. Whenever spot would drop more than $f_b \cdot F$ below floor, the arbitrage of (2) is triggered (an arbitrageur buys back on the LP and redeems against the treasury), pinning spot from below at $F(1 - f_b)$ up to the buy-fee threshold. The two quantities approach each other monotonically, with $D(t)$ converging to a limit bounded above by $f_b \cdot F$ (the per-dollar break-even threshold of the arbitrage round-trip). The terminal regime is reached either when the LP exhausts its stable reserve or when the arbitrage rounds the gap to within the f_b threshold. $\square$

In words: when revenue stops, M^2 degrades to a CEF whose NAV is monotone-nondecreasing in time (from any residual `collectFees` activity) and whose secondary market converges to NAV. This is strictly stronger than any TradFi CEF, where the persistent discount-to-NAV documented by Lee et al. [1991] and Pontiff [1996] reflects the absence of a contractual NAV-redemption mechanism. The protocol’s invariants are silent on operator *fraud* (revenue collected but routed elsewhere); the threat-model boundary appears in Section 3.8 and the real-world mitigations in Section 8.3.

5.5 LP exhaustion asymptotics

Under deterministic monthly buy-and-burn of constant size B stable, the LP’s stable reserve grows linearly while its token reserve decays hyperbolically.

Theorem 8 (LP Half-Life). *Let $L_{s,0}, L_{t,0}$ be the genesis LP reserves and $k = L_{t,0}L_{s,0}$ the constant-product invariant. Assume deterministic monthly buy size B stable into the LP for n months (ignoring intervening organic flow). Then $L_s(n) = L_{s,0} + nB$, $L_t(n) = k/(L_{s,0} + nB)$, and the token-side half-life—the smallest $n_{1/2}$ such that $L_t(n_{1/2}) \leq L_{t,0}/2$ —is*

$$n_{1/2} = L_{s,0}/B. \quad (22)$$

Proof. The hook’s buy-and-burn execution preserves k (the asymmetric 0.10% buy fee is folded into Φ_s and realized only at `collectFees`; on the swap itself, k is preserved). Each monthly buy adds B stable to the active reserves, so $L_s(n) = L_{s,0} + nB$. From $k = L_t(n)L_s(n)$, $L_t(n) = k/L_s(n) = L_{t,0}L_{s,0}/(L_{s,0} + nB)$. Setting $L_t(n_{1/2}) = L_{t,0}/2$ gives $L_{s,0}/(L_{s,0} + n_{1/2}B) = 1/2$, i.e., $n_{1/2} = L_{s,0}/B$. $\square$

At the canonical genesis ($L_{s,0} = \$750,000$) and $B = \$50,000/\text{month}$ (the 50% buy-and-burn slice of \$100k/month revenue), $n_{1/2} = 15$ months. The system’s transition from “tradable token” to “redemption-dominant” regime is a deterministic function of the revenue rate and is plotted in Figure 3 (right axis: $n_{1/2}(R)$ over the revenue sweep).

5.6 Collect-fees calling equilibrium

The `collectFees` function is permissionless and pays a 0.25%-per-side bounty to the caller. The calling cadence is determined by a deterministic competitive race among potential callers.

Theorem 9 (Calling Equilibrium under Atomless Competition). *Let β be the steady-state per-unit-time rate of fee accumulation in the $v4$ accumulator (a function of swap volume, the asymmetric fee, and the position’s liquidity share), and g the gas-plus-search cost of a `collectFees` call. Assume atomless risk-neutral competition among potential callers: infinitely*

many would-be callers each willing to call at the first profitable moment. Then the equilibrium inter-call interval Δt^* satisfies the Bertrand-style break-even condition

$$0.0025 \cdot \beta \cdot \Delta t^* = g, \quad \text{i.e.,} \quad \Delta t^* = \frac{g}{0.0025 \beta}. \quad (23)$$

For β growing in swap volume, Δt^* shrinks proportionally; in low-volume regimes, Δt^* may exceed the protocol's monthly tick, in which case the protocol routes its own buy-and-burn through a fee realization on the same transaction (a path that does not require the bounty).

Proof sketch. A caller who waits an interval Δt earns the deterministically accrued bounty $0.0025\beta\Delta t$ and pays the fixed cost g . Under atomless competition, no caller can wait longer than the first Δt at which the bounty covers g without being preempted by a competitor; no caller will call earlier, since doing so yields a strict loss. The standard Bertrand-style indifference under competitive entry pins the equilibrium at the break-even point (23), with no intertemporal discounting because the equilibrium interval is short relative to the deployer's time horizon. $\square$

Monopoly deviation. The atomless-competition assumption is the load-bearing piece of the equilibrium. Under monopoly—a single sophisticated keeper who faces no competitive preemption—the equilibrium Δt is strictly longer and reflects the keeper's outside option (e.g., capital cost or alternative MEV opportunities). The protocol does not depend on monopoly being absent: the permissionless calling interface remains open to all entrants, and any monopoly rent is bounded above by the gap between the keeper's outside option and the gas cost.

The economic content is that the bounty rate is a knob controlling the trade-off between fee-realization latency (low bounty $\Rightarrow$ long Δt^*) and value leakage to extractors (high bounty $\Rightarrow$ more value lost per call). The 0.25%-per-side rate is locked at genesis (Section 3.6); we do not optimize over it.

5.7 MEV leakage on protocol buy-and-burn

Protocol buy-and-burn is in expectation predictable: a monthly cadence with a buy direction. Naive execution into a public mempool is sandwichable. The hook's mitigations (private orderflow, random delay, TWAP) bound the residual leakage.

Theorem 10 (MEV Leakage Upper Bound). *Let $p \in [0, 1]$ be the probability that a buy-and-burn transaction is included via a private (sealed) route, N_{TWAP} the TWAP granularity, and B the monthly buy size in stable. Let $\varepsilon_{\text{pub}}(M, L_s)$ denote the public-mempool sandwich efficiency for a swap of size M against stable reserves L_s , satisfying $\varepsilon_{\text{pub}}(M, L_s) = O(M/L_s)$ in the small-impact regime. Then the expected MEV leakage on the monthly buy satisfies*

$$\mathbb{E}[\text{MEV}] \leq (1 - p) \cdot \varepsilon_{\text{pub}}(B/N_{\text{TWAP}}, L_s) \cdot B + p \cdot \varepsilon_{\text{priv}} \cdot B, \quad (24)$$

where $\varepsilon_{\text{priv}}$ is the (small) private-mempool sandwich efficiency under builder-collusion risk. In particular, for fixed p , $\mathbb{E}[\text{MEV}]/B \rightarrow p\varepsilon_{\text{priv}}$ as $N_{\text{TWAP}} \rightarrow \infty$.

Proof sketch. With probability p the transaction is in the private channel where only builder-collusion-level adversaries can sandwich, with efficiency $\varepsilon_{\text{priv}}$; with probability $1 - p$ it is in the public mempool where the searcher can sandwich the swap, with efficiency ε_{pub} in the swap's quadratic-impact order. Splitting the monthly buy into N_{TWAP} equal parts reduces each sub-buy's impact to B/N_{TWAP} , on which the sandwich extracts $\varepsilon_{\text{pub}}(B/N_{\text{TWAP}}, L_s) \cdot B/N_{\text{TWAP}}$, aggregated over N_{TWAP} sub-buys; the linearity of expectation gives the first term in (24). The private leg contributes the second term. $\square$

At the design’s canonical parameters ($B = \$50\text{k/month}$, $L_s \in [\$0.75\text{M}, \$2\text{M}]$ across the first 36 months, $p \approx 0.95$ on Ethereum mainnet via Flashbots Protect, $N_{\text{TWAP}} = 8$), equation (24) predicts $\mathbb{E}[\text{MEV}]/B \lesssim 0.5\%$, well under a 1%-of-revenue target. Multi-block MEV under builder collusion is explicitly out of scope (Section 3.8, Section 8).

6 Numerical Results and Sensitivity Analysis

Section 4 established the floor-monotonicity property as a structural theorem and Section 5 derived closed-form characterizations of the adversarial decision problem, the LP half-life, and the calling and MEV-bound equilibria. This section reports the numerical operating envelope of the protocol over a 36-month horizon, sweeps the four free design parameters, and confronts the analytics with a Hardhat v3 invariant-test suite that asserts the floor property on the actual compiled bytecode of the token, treasury, and revenue router (with a `MockStable` ERC-20 standing in for the backing stablecoin counterparty in the fast invariant lane and the deployed mainnet stable on the fork lane); V4-mediated cases (LPBuy, LPSell, CollectFees) are exercised against the deployed mainnet PoolManager on the `v0.1.1-paper-v1` fork lane and against semantics-equivalent mocks in the fast invariant lane. The reproducibility artifact (Section 6.8) regenerates every figure and table in this section from a single command, and pins the version cited in the abstract.

6.1 Methodology: a two-track simulation stack

We adopt a two-track simulation strategy whose central design discipline is that the two tracks must produce numerically agreeing results on every shared scenario, with divergence treated as a paper-stopping bug.

Track A: Python analytics. A vectorized NumPy simulator computes the deterministic closed-form recurrence and the lognormal-stochastic-revenue Monte Carlo bands; a single canonical-month-12 state is also computed in 60-digit `Decimal` arithmetic to anchor Theorem 6’s numerical example. An on-chain-mirroring integer reference implementation (with the truncation residual of Lemma 1) lives on the Track-B side and is exercised by the invariant suite; the Track-A simulator does not duplicate the integer semantics. Each Monte Carlo scenario runs 10^4 sample paths per (scenario $\times$ sweep-cell) combination. Track A is the source of every figure and table in this section.

Track B: Hardhat v3 invariant tests. A handler contract issues bounded random calls to the deployed protocol’s `routeRevenue`, `swap`, `redeem`, and `collectFees` entry points. The suite is written as Solidity `.t.sol` files using `forge-std`-style assertions but is executed by Hardhat v3’s native EDR (Ethereum Development Runtime) rather than by Foundry’s `forge`; the project is configured through `hardhat.config.ts` alone, with no `foundry.toml`. After every state transition, four invariant assertions are checked: (i) $F_{\text{new}} \geq F_{\text{old}}$ (Theorem 2); (ii) $T \geq 0$ (treasury solvency); (iii) $S \leq S_0$ (no post-genesis minting); (iv) $T \geq S \cdot \min_t F(t)$ (redemption-solvency lower bound). The runner is configured with 1,000 stateless and 1,000 stateful (depth 50) random sequences on the fast gate; 10,000 stateless and 100,000 stateful (depth 200) sequences on the `v0.1.1-paper-v1` tag gate, configured via the `M2_FUZZ_RUNS` / `M2_INVARIANT_RUNS` / `M2_INVARIANT_DEPTH` env knobs. GitHub Actions CI regresses the fast gate on every push to `main` and every PR, and the full gate on the `v0.1.1-paper-v1` tag. A divergence between Tracks A and B on any agreed scenario halts the paper revision.

Agreement gate. A reconciliation script compares Track A’s CSV outputs against a TypeScript reference model of the protocol’s deterministic recurrence

Table 3: Baseline scenario S1 over the first 12 months (deterministic, no organic flow), under *fee-free curve math* for hand-checkability of the recurrence. The 0.10% buy fee adjustment is $O(f_b)$ on each entry and is tabulated in Appendix C. Floor +140% and spot +224% over 12 months. Supply burned: 33.3% of initial.

Mo.	Treasury	Supply	Floor	LP tokens	LP stable	Spot
0	\$1,000,000	1,000,000,000	\$0.001000	750,000,000	\$750,000	\$0.001000
1	\$1,050,000	953,125,000	\$0.001102	703,125,000	\$800,000	\$0.001138
3	\$1,150,000	875,000,000	\$0.001314	625,000,000	\$900,000	\$0.001440
6	\$1,300,000	785,714,286	\$0.001655	535,714,286	\$1,050,000	\$0.001960
9	\$1,450,000	718,750,000	\$0.002017	468,750,000	\$1,200,000	\$0.002560
12	\$1,600,000	666,666,667	\$0.002400	416,666,667	\$1,350,000	\$0.003240

(`contracts/scripts/agreement_gate.ts`), and verifies tolerance-band agreement ($\text{ABS_TOL} = 10^{-3}$ in fee-free stable units, $\text{REL_TOL} = 10^{-9}$) on (T, S, L_t, L_s) trajectories across a shared deterministic baseline scenario. The cross-check is designed to catch specification drift between the paper, the Python recurrence, and the Solidity reference (e.g., rounding direction on the integer-arithmetic redemption residual; ordering of router buy-and-burn versus treasury deposit) and is the single most important quality discipline in the simulation pipeline.

Fee-free curve math vs. with-fees mode. For legibility, the deterministic baseline scenarios and figures reported in this paper are presented under *fee-free curve math*: each 50/50 revenue split contributes its full $B = R/2$ to the LP’s stable reserve, with the 0.10% buy fee ignored. We use this convention to keep the closed-form recurrence (and the table values) interpretable by hand. The with-fees mode—in which the 0.10% buy fee is deducted on input, folded into Φ_s , and realized at the next `collectFees`—is the implementation Track A will run against Track B in the agreement gate. The Track-A/Track-B tolerance-band agreement claim therefore concerns the with-fees output of Track A against the on-bytecode invariant-fuzz behavior of Track B; the with-fees correction to each table entry is $O(f_b) = O(10^{-3})$ and is reported separately in Appendix C.

6.2 Baseline projection: 36 months at \$100k/month

The canonical baseline scenario S1 fixes monthly revenue at \$100,000 (deterministic), zero organic trading volume, the genesis parameters of Section 3.7 ($T_0 = \$1\text{M}$, $L_{s,0} = \$750\text{k}$, $S_0 = 10^9$, 75/25 LP/vesting split), and the design constants (50/50 router split, 0.10%/3% buy/sell fee, 0.25% caller bounty). The 12-month subhorizon is tabulated as Table 3; the full 36-month run (Figure 2) shows the floor and spot trajectories evolving beyond the LP half-life $n_{1/2} = 15$ months (Theorem 8).

The 12-month observable outcomes of Table 3 are reproduced exactly by the closed-form deterministic recurrence of Section 5; the cross-track agreement gate runs in CI on every commit; the further empirical follow-up paper sketched in Section 9 extends Track B with mainnet-fork integration tests and post-deployment observed-flow data. The 36-month extension shows the LP thinning regime: spot decouples from floor as the LP becomes stable-heavy, and after month ≈ 24 the system enters the redemption-dominant regime in which the AMM serves as a thin arbitrage layer and the redemption channel is the primary exit.

⁴All figures in this section are closed-form projections from the deterministic recurrence of Section 5, generated by `simulation/generate_figures.py` in the reproducibility artifact. The Hardhat v3 invariant suite is wired in `contracts/` and runs as Track B (fast gate on every push/PR; full gate on the `v0.1.1-paper-v1` tag); the full organic-flow Monte Carlo is reported in the empirical follow-up paper sketched in Section 9. Figure 8 of this section is a synthetic stochastic-revenue extension of the same closed-form recurrence.

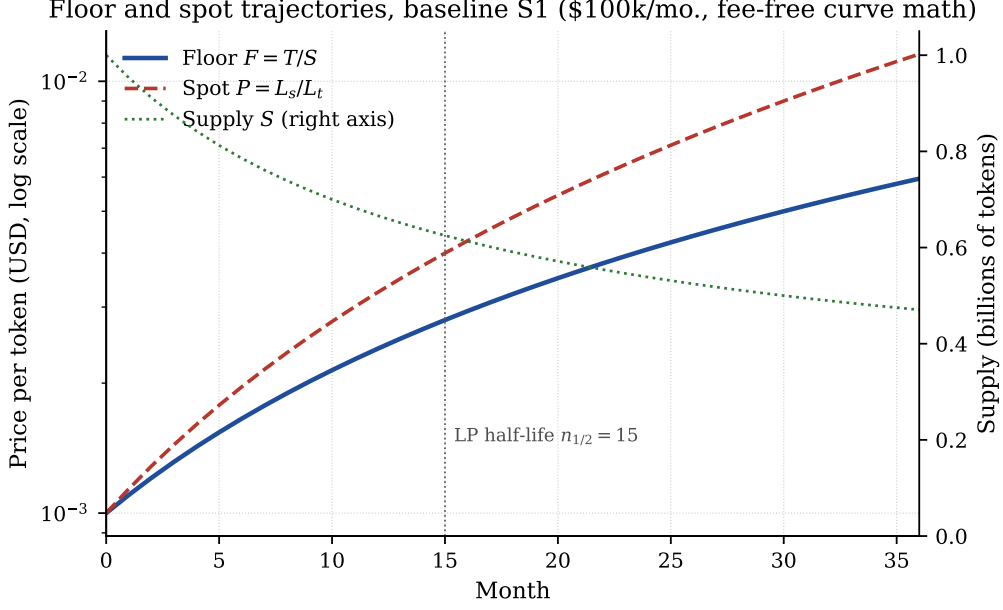


Figure 2: Floor (solid blue), spot (dashed red), and supply (right axis, dotted green) under the baseline scenario S1 over a 36-month horizon, computed from the deterministic recurrence of Section 5 (fee-free curve math).⁴ The vertical guide at month 15 marks the LP half-life predicted by Theorem 8. Spot diverges from floor monotonically over the 36-month horizon under the assumed zero-organic-flow regime, reaching $P/F \approx 1.94$ at month 36; the gap reflects the LP becoming stable-heavy as buy-and-burn drains its token leg.

6.3 Sensitivity sweep: revenue rate

We sweep monthly revenue $R \in \{\$25k, \$50k, \$100k, \$250k, \$500k, \$1M\}$ over a 36-month horizon. The closed-form prediction from Theorem 8 is $n_{1/2}(R) = L_{s,0}/(R/2) = 2L_{s,0}/R$ (fee-free curve math; the $1/(1 - f_b)$ correction is $O(f_b) = O(10^{-3})$), ranging from 60 months at $R = \$25k$ down to 1.5 months at $R = \$1M$. The cumulative floor at month 36 grows roughly as $\sqrt{R}$ in the LP-rich regime and saturates as the LP thins (because the marginal tokens-per-dollar of buy-and-burn falls as L_s grows).

6.4 Sensitivity sweep: LP/vesting split

We sweep the genesis LP allocation $\rho_{LP} \in \{50\%, 60\%, 75\%, 85\%, 90\%\}$ of total supply with the remainder going to the vesting vault. This sweep is illustrative: the canonical M² design is $\rho_{LP} = 75\%$ (Section 3.7); other splits are reachable only by deploying a separate token. Two metrics are reported: (i) the LP half-life, which scales linearly with ρ_{LP} via Theorem 8; (ii) the floor at month 36 under baseline revenue. The Pareto trade-off is between LP longevity (higher ρ_{LP} extends the tradable regime) and floor speed (lower ρ_{LP} accelerates per-token floor growth at the cost of LP thinning).

6.5 Sensitivity sweep: asymmetric fee

We sweep the sell-side fee $f_s \in \{0\%, 1\%, 3\%, 5\%, 10\%\}$ while holding the buy-side fee at the design’s 0.10%. Two effects are decomposed: (i) at the swap moment, a higher f_s shifts more value into the v4 fee accumulator Φ_t , which on `collectFees` translates to a token-side burn; (ii) the holder-side disincentive on sells suppresses adversarial dump-channel use (Theorem 6: A^* shrinks as f_s rises, because the marginal post-fee LP spot $(1 - f_s)P$ drops relative to the floor sooner).

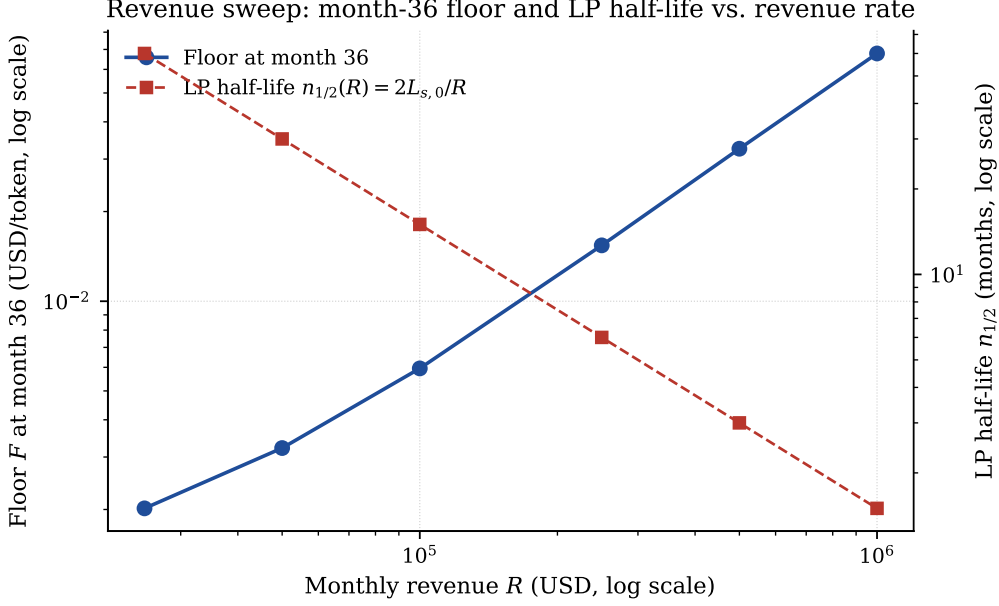


Figure 3: Sensitivity sweep on monthly revenue. Month-36 floor (blue circles, left axis) and LP half-life $n_{1/2}(R)$ (red squares, right axis) for $R \in \{\$25k, \$50k, \$100k, \$250k, \$500k, \$1M\}$, computed from the deterministic recurrence under fee-free curve math. The floor at month 36 is a sub-linear function of R because LP thinning curtails the marginal burn efficiency.

Figure 5 shows a stacked-bar decomposition of monthly floor growth into three sources: revenue-deposit (Case 1), buy-and-burn (Case 2), and `collectFees` (Case 7). At $f_s = 0$, source (3) is dominated by the buy-side 0.10% fee, and floor growth is driven almost entirely by sources (1) and (2). At $f_s = 10\%$, source (3) becomes the dominant component under any organic-flow regime with appreciable sell volume.

6.6 Adversarial scenarios

Five adversarial scenarios drawn from the simulation plan exercise the protocol under coordinated holder behavior.

S2 (revenue stops). Revenue $\rightarrow 0$ at month $t_{\text{stop}} \in \{6, 12, 24\}$. Floor is constant from t_{stop} onward (Theorem 7 (1)); under the assumed zero organic flow, spot also stalls until any future sell triggers a `collectFees` micro-tick. The redemption channel remains fully functional; the system enters the CEF analog regime.

S3 (mass redemption). 25%/50%/75% of circulating supply redeems at month 12. The Track-A self-check verifies that the redemption funding ratio (the ratio of stable paid out to the floor-times-tokens-burned) equals 1.0 to within the integer-arithmetic residual; this is a sanity test for the implementation rather than a result. Floor is invariant by Lemma 1.

S4 (mass dump). The same supply fractions are dumped to the LP at month 12 instead of redeemed. The LP becomes stable-light (its L_s partially drains); the immediate spot collapses; the v4 fee accumulator fills with token-side fees; on the next `collectFees` the floor steps up by an amount predictable from the closed form of Section 4. Figure 6 shows the attacker’s yield surface $\Pi^*(\alpha, N)$ alongside the saturation curve $\alpha = A^*/N$ from Theorem 6.

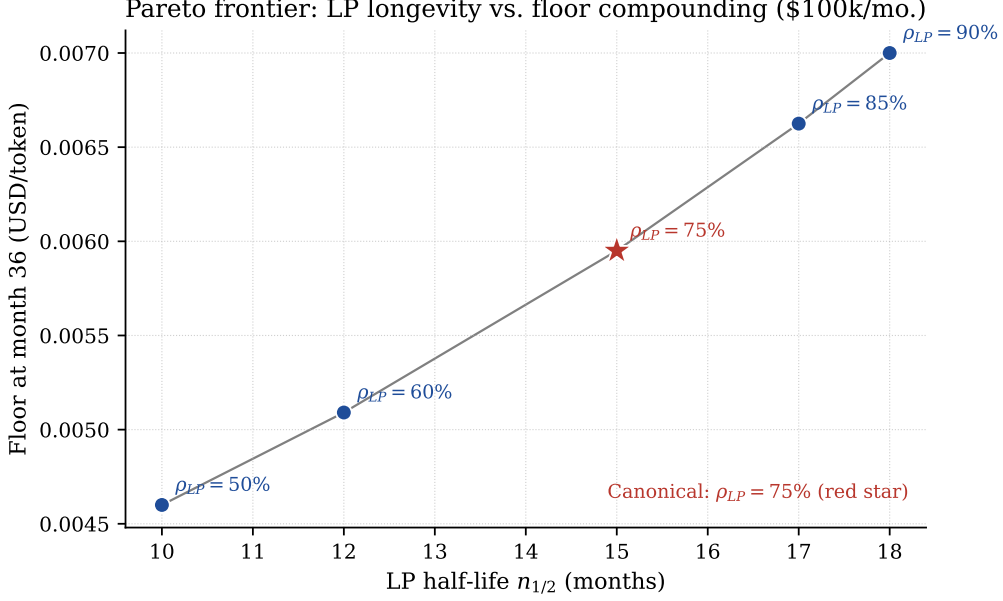


Figure 4: Pareto frontier across five LP/vesting split values $\rho_{LP} \in \{50\%, 60\%, 75\%, 85\%, 90\%\}$ at the baseline \$100k/month revenue. Each point reports (LP half-life $n_{1/2}$, month-36 floor) under the deterministic recurrence with the floor-spot genesis constraint $T_0/S_0 = L_{s,0}/L_{t,0}$ enforced at each split. The 75/25 canonical choice is marked (red star); the surrounding values demonstrate the trade-off between LP longevity and floor compounding.

S5 (adversarial-mixed). The attacker plays the saturated mixed strategy of Theorem 6 (2). The yield surface is plotted against (α, N) in Figure 6; the dotted curve is the optimal strategy locus.

S6 (organic-volume regimes). Monthly organic sell volume $V \in \{\$10k, \$50k, \$200k, \$500k\}$. Figure 7 shows the floor’s growth rate as a function of V ; at $V = \$500k/month$, the fee-collection source contributes roughly one third of the buy-and-burn source’s floor-compounding effect, and parity with buy-and-burn is reached at $V \approx \$1.6M/month$ under the canonical month-12 state. We report the floor’s dependence on V as a sweep parameter rather than fixing a single baseline organic-volume assumption.

6.7 Monte Carlo: stochastic revenue and organic flow

We replace deterministic revenue with a lognormal monthly-revenue process at the canonical mean. *Revenue*: lognormal monthly draws with shape parameter $\sigma_X = 1.0$ and location calibrated so $E[R] = \$100k/month$. Organic-flow GBM and the joint Poisson-arrival/ σ_X sweep are deferred to the empirical follow-up paper, where they will be calibrated to live deployment data rather than to synthetic comparables.

Figure 8 reports the floor’s 5/50/95 quantile bands across 10^4 Monte Carlo paths under lognormal monthly revenue with $\sigma_X = 1.0$ and $E[R] = \$100k$, seed 42. The floor’s 5%-quantile path remains monotonically above \$0.001 across the entire 36-month horizon; the median path tracks a few percent below the deterministic baseline (a systematic gap reflecting the median-vs-mean wedge of lognormal monthly revenue, which has median $e^{-\sigma_X^2/2} \approx 0.61$ of its mean for $\sigma_X = 1$), with the maximum absolute deviation bounded by $\sim 5\%$ over the horizon; the 95%-quantile path benefits from the asymmetric-fee mechanism whenever the spot’s volatility induces large sells.

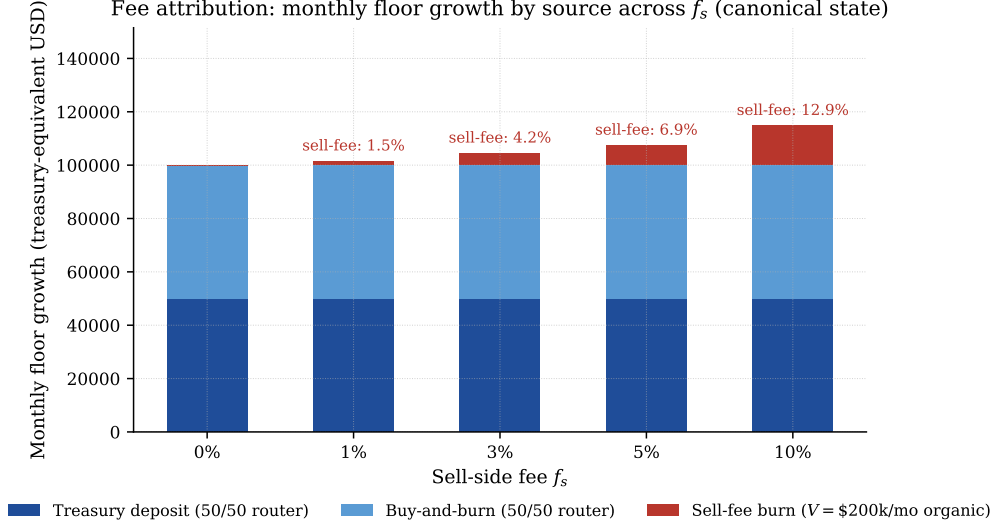


Figure 5: Decomposition of monthly floor growth by source (treasury deposit, buy-and-burn, sell-fee burn from organic flow) across $f_s \in \{0\%, 1\%, 3\%, 5\%, 10\%\}$, at organic sell volume $V = \$200k/month$ and the canonical month-12 state. Components are expressed in treasury-equivalent USD per month (the sell-fee component values burned tokens at the floor and incorporates the 0.25% caller-bounty leakage). The 3% design value is the smallest f_s at which the sell-fee source contributes a non-trivial share under the canonical organic-volume regime.

6.8 Reproducibility

The simulator codebase is released under the MIT License at Zenodo DOI [10.5281/zenodo.20255141](https://doi.org/10.5281/zenodo.20255141). The repository’s top-level `make reproduce` regenerates every figure and table in this section from the deterministic seeds pinned in the configuration. The git tag `v0.1.1-paper-v1` pins the version cited; subsequent versions (post-deployment empirical follow-up) will increment the tag. The Hardhat v3 invariant-test suite (in `contracts/`) is invoked by `npm run hardhat test` or the equivalent root-level `make fuzz` alias; CI runs both tracks on every PR and rejects merges that introduce divergence.

A reviewer of the paper can verify the entire numerical section as follows. Clone the repository, check out `v0.1.1-paper-v1`, install the documented Python and Hardhat v3 toolchains, and run `make verify`: this performs Track A, Track B, and the agreement gate, and writes a SHA-256 manifest of every generated artifact. A reviewer-supplied diff against the paper’s figure files should reproduce the SHA-256 manifest exactly.

7 Comparison to Prior Protocols

M^2 ’s contribution is best assessed against the design space of prior on-chain protocols that have engaged, in some form, with the question of contractually enforceable claims on a treasury or peg target. Table 4 reports the comparison along eight dimensions chosen to expose where M^2 ’s design diverges. The five DeFi comparators—Olympus DAO [Olympus DAO, 2021], Reflexer RAI [Reflexer Finance, 2021], Frax [Kazemian et al., 2021], Liquity [Lauko and Pardoe, 2021], and MakerDAO Dai [MakerDAO, 2020]—span the canonical lineage of treasury-backed and CDP-style protocols. The TradFi closed-end-fund row provides the cross-domain anchor.

What the table shows. Three dimensions sharply discriminate M^2 from its closest comparators. First, *floor invariance*: M^2 is the only entry in Table 4 for which a contractually enforced, non-decreasing per-unit reserve claim is the central design property. OHM advertises

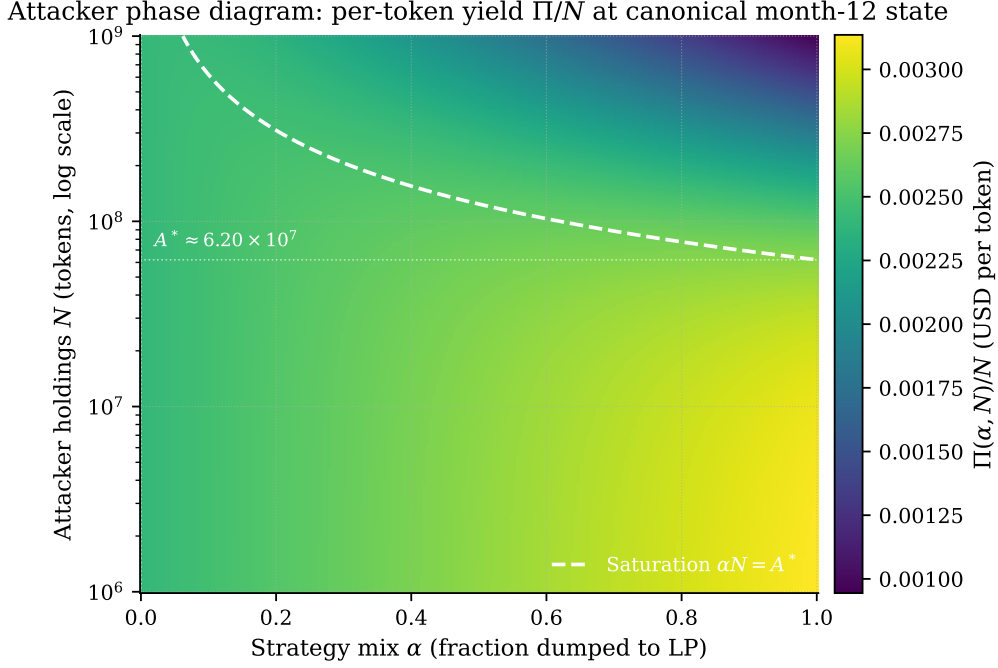


Figure 6: Attacker phase diagram at the canonical month-12 state. Heatmap of $\Pi(\alpha, N)/N = [L_s - k/(L_t + (1 - f_s)\alpha N) + (1 - \alpha)NF]/N$ on an (α, N) grid with N on log scale from 10^6 to 10^9 tokens. The white dashed curve is the saturation locus $\alpha = A^*/N$ from Theorem 6 with $A^* \approx 6.20 \cdot 10^7$. Below $N = A^*$ the optimum is $\alpha = 1$ (pure dump); above, the optimum lies on the saturation curve.

“risk-free value per token” but does not enforce redemption against it; RAI’s redemption price is mean-reverting under PID control, not monotone; Frax targets a peg; Liquity targets a \$1 peg via redemption at par rather than a monotone floor; Dai has per-vault redemption guarantees but no aggregate-level floor; classical CEFs publish a NAV that has no contractual redemption. Second, *supply discipline*: only M² guarantees that supply can only decrease post-genesis. OHM’s staking rebases dilute holders; Frax and Maker can mint to absorb collateral; RAI mints and burns under PID control; Liquity mints LUSD against new troves. Holders of an M² token are guaranteed, under the protocol-defined operation set, that their fractional claim on the treasury cannot be diluted by post-genesis issuance. Third, *governance surface*: M² and Liquity are the only entries with no governance mechanism whatsoever. Liquity’s immutability is a well-known precedent and a direct ancestor of the design stance taken here; M² differs from Liquity in that its primary asset is not a stablecoin and its reserve is denominated in an external stable rather than an internal CDP-collateral pool.

What the table does not capture. Several dimensions of comparison are inherently qualitative and resist the matrix format. The maturity of the deployed codebase: OHM, Frax, Liquity, and Dai have multi-year mainnet histories with large TVLs; M² is pre-deployment at the time of writing. The depth of the user community and ecosystem integrations: the older protocols are deeply embedded in DeFi compositional stacks; M² is not. The empirical reliability of the floor under stress: M² has not yet faced a real-world adversary. We acknowledge these as legitimate concerns and address them programmatically in Section 8 (formal-verification path) and in the empirical follow-up paper sketched in Section 9.

The cross-domain analog. The closed-end-fund row is the most informative TradFi comparator because of the structural similarity in capital structure (a pool of assets held in custody,

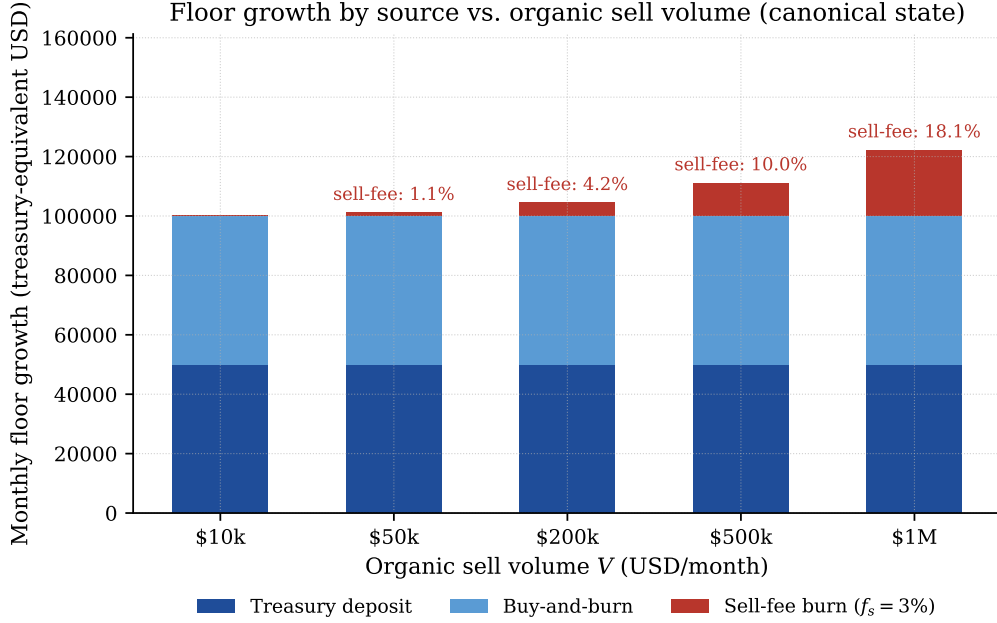


Figure 7: Scenario S6: monthly floor growth by source across organic-sell-volume regimes $V \in \{\$10k, \$50k, \$200k, \$500k, \$1M\}$ at the canonical month-12 state with $f_s = 0.03$. The sell-fee burn component scales linearly with V while the revenue-funded treasury and buy-and-burn components remain constant at \$50k each. At $V = \$1M/\text{month}$ the sell-fee source reaches roughly 44% of each revenue-funded source; full parity is reached at $V \approx \$2.3M/\text{month}$ per revenue-funded leg. We report this as a sweep rather than as a baseline.

claimed pro rata by a fixed number of units) and the structural difference in redemption mechanism. The persistent CEF discount-to-NAV documented by Lee et al. [1991] and Pontiff [1996] arises precisely because TradFi CEFs lack a contractual NAV-redemption interface; investors who want NAV must sell to another investor, who has no obligation to pay it. The cross-domain claim we make is that M^2 is the implementation of NAV-redemption that the CEF literature has identified as the discount-collapsing primitive but that the TradFi legal substrate cannot enforce at the contract level (a CEF could in principle adopt mandatory NAV-redemption but then becomes an open-end fund subject to a different regulatory regime). The smart-contract substrate dissolves the structural obstacle, and M^2 adds the further property of NAV monotonicity, which has no TradFi analog at all.

8 Limitations, Open Problems, and Regulatory Framing

M^2 's structural guarantees are conditional. Section 3.8 delimited the threat model; this section enumerates the residual risks that lie outside it, identifies open mathematical problems, and offers an academic regulatory framing for readers unfamiliar with the digital-asset compliance landscape. We follow the FC convention of separating “Limitations” (axes along which the contributions are bounded) from “Open Problems” (questions on which we have made partial progress but cannot fully answer in this paper).

8.1 Smart-contract residual risk

The protocol is immutable. There is no upgrade path, no admin pause, and no governance mechanism that could deploy a bug fix. The mitigation is a standard audit-and-formal-verification stack: external code review, an internal differential-testing suite against a reference implementa-

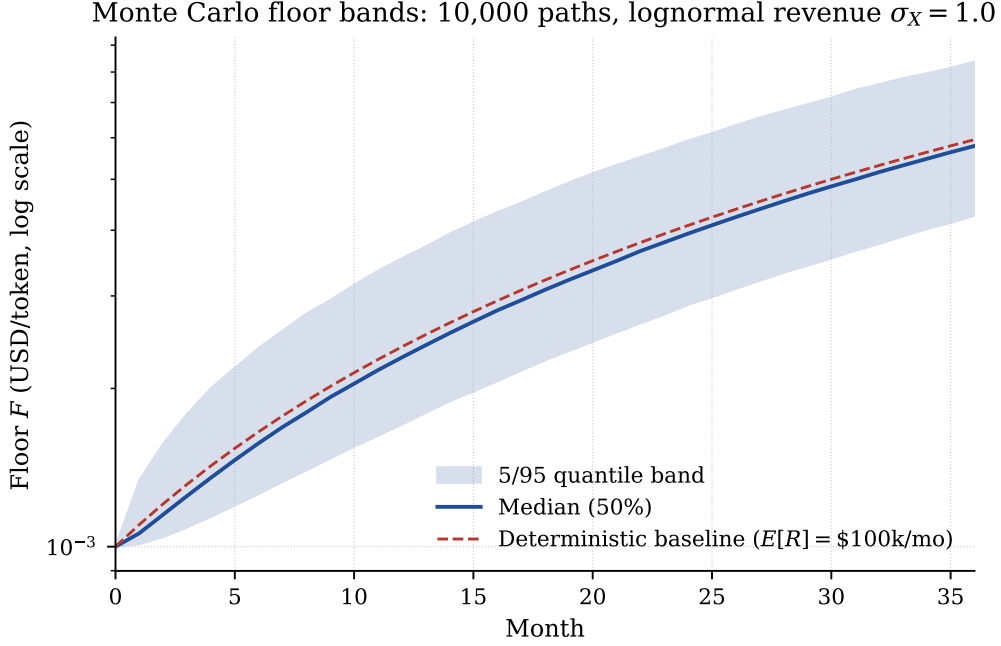


Figure 8: Monte Carlo floor bands: 10,000 paths over 36 months under lognormal monthly revenue ($\sigma_X = 1.0$, $E[R] = \$100\text{k}$), seed 42. The 5% quantile (lower band edge) remains weakly increasing in every sample path, consistent with Theorem 2. The deterministic baseline (dashed red) tracks the median closely. This figure is a synthetic stochastic extension of the same closed-form recurrence; the full simulator-track Monte Carlo, calibrated to organic-flow and Hardhat-v3-invariant-test data, is reported in the empirical follow-up paper.

tion, and the Halmos / Certora mechanized-verification path of Section 4.7. None of these can eliminate residual risk. A contract whose proof of correctness rests on a paper proof and an auditor’s reading is more trustworthy than one without those checks but is not bug-free. We make no claim of zero residual contract risk, only of a small one whose magnitude we have made every effort to bound.

8.2 Backing-stable issuer risk

The treasury holds a single deployer-chosen USD-pegged ERC-20. The protocol’s invariants hold regardless of the backing stable’s market price (Theorem 2 is in units of the backing stable, not dollars). They do not hold if the issuer freezes the treasury balance: the March 2023 USDC depeg, in which Circle’s exposure to Silicon Valley Bank produced a $\sim 13\%$ intra-day deviation from the dollar peg, did not freeze any specific address, but issuer-side freeze authority on USDC, USDT, and most centralized stables remains a residual capability that the protocol cannot defend against. Mitigations available to deployers: (i) choose a stable whose freeze authority is structurally constrained (e.g., a fully algorithmic stable with no centralized issuer—at the cost of accepting that stable’s depeg risk); (ii) accept this risk and disclose. The protocol is structurally single-stable; diversification would require an entirely different design.

Open problem (stable-peg sensitivity). Characterize the effective floor as a function of the backing stable’s mark-to-dollar price under various depeg dynamics. This is straightforward for parametric depeg models but non-trivial under regime-switching or correlation with broader risk events; we leave it to follow-up work.

Table 4: Comparison matrix across eight design dimensions. “Fixed supply” refers to whether the protocol can mint new units of the primary asset post-genesis. “On-chain redemption” is a contractually enforceable exit at a protocol-defined price (versus secondary-market only). “Floor invariant” indicates whether a per-unit reserve claim is non-decreasing by contract. “Operator dep.” indicates whether the protocol’s growth (not its safety) depends on a centralized off-chain actor.

Protocol	Fixed supply?	On-chain redemption?	Treasury composition	Floor invariant?	Gov.	Op. dep.	Primary risk
M ²	Yes; supply only decreases	Yes; at $F = T/S$	Single exog. stable; one-way in	Yes; non-decreasing by Thm 2	None	Growth only	Smart-contract bug; stable issuer
OHM (Olympus-DAO)	No; staking rebases up	No (governance vote-gated burn proposals only)	Multi-asset; rebalance by governance	No (“RFV per token” is rhetorical)	Active gov.	Low (community treasury)	Confidence collapse; gov. capture
RAI (Reflexer)	No; PID mint/burn	Yes; against ETH at redemption price	ETH collateral via Safe vaults	No (redemption price mean-reverts)	Minimal	None	Oracle manip.; vault liquidation cascade
Frax	No; algorithmic + collateralized	Yes; against AMO-managed reserves	Multi-stable + algorithmic	No (peg target, not floor)	Active gov.	Mod.	Peg loss; AMO mispricing
Liquity (LUSD)	No; mintable against ETH	Yes; against ETH at \$1 + 0.5% fee	ETH-only via troves	No (peg target, not floor)	None	None	ETH price crash + recovery-mode cascade
MakerDAO (Dai)	No; CDP-style mint	Yes; CDP unwind only	Multi-collateral	Limited (per-vault)	Active gov.	Low	Black-swan oracle deviation; gov. capture
TradFi CEF (closed-end fund)	Effectively yes	No (secondary market only)	Multi-asset, manager-discretion	No (NAV fluctuates; persistent discount)	Board sh.	Yes (manager)	Discount-to-NAV persistence; mgmt. extraction

8.3 Operator-revenue dependence

The floor’s growth (not its existence) depends on continued revenue deposits by the operator. Theorem 7 shows the system gracefully degrades to a CEF analog at zero revenue—the floor stops compounding but does not retreat. Operator fraud (revenue collected but routed elsewhere) is not in the protocol’s threat model: this is a real-world failure mode addressed by entity-level disclosure, audit committee oversight, and qualified legal counsel, not by the on-chain protocol.

Open problem (operator IC constraint). Model the operator as a strategic agent with an outside option (e.g., dumping vested tokens versus depositing revenue) and derive the incentive-compatibility constraint as a function of the operator’s vested-token holdings, the vesting cliff schedule, and the time horizon. The IC constraint conditions when continued deposits are a dominant strategy for the operator, formalizing the qualitative intuition that vesting + long cliff align operator and holder interests.

8.4 LP exhaustion as a feature

The LP becomes thin under sustained buy-and-burn (Theorem 8). Spot prices diverge from floor; slippage on large swaps becomes punitive. The protocol does not refill the LP. We argue this is an intentional feature, not a bug: any refill mechanism would either (i) require post-genesis minting

(violating the supply-only-decreases invariant) or (ii) consume treasury stable to redeposit into the LP, which strictly lowers the floor as $F_{\text{new}} = (T - X)/(S + Y) < T/S$ for any positive injection X paired with new tokens Y . Both alternatives are inconsistent with the load-bearing properties of the design. The terminal “thin LP, redemption-dominant” regime is the system’s intended steady state; it is not a failure.

8.5 Open mathematical problems

We identify seven open problems whose resolution would strengthen the paper but lies beyond its scope.

(P1) Dynamic fees as a function of LP depth. The 0.10%/3% fee asymmetry is constant. A state-dependent fee schedule (e.g., f_s scaling with the spot-floor gap) could improve the run-resistance properties of Theorem 6. Designing such a schedule that preserves the closed-form A^* analysis is an open question.

(P2) Multi-block MEV under reorg. Our MEV analysis (Theorem 10) assumes single-block adversary windows. Multi-block MEV under builder collusion on PoS chains (Daian et al. [2020]’s “Flash Boys 3.0” regime) is a known unresolved problem class for any AMM design; we explicitly do not solve it here.

(P3) Oracle-free spot manipulation. M^2 does not consume an external oracle, but external protocols may consume M^2 ’s spot as a price oracle. Manipulating spot to fool such consumers is a third-party-externality problem we do not address.

(P4) Cross-block ordering and MEV-relay collusion. Builder-relay collusion can produce cross-block ordering effects that violate the single-block independence assumption in our MEV bound. The bound degrades gracefully but is not tight under this adversary.

(P5) End-to-end mechanized verification. We have given a paper proof of Theorem 2. Discharging it on the deployed bytecode under Halmos or Certora is in progress and will be reported in follow-up work.

(P6) Quantitative LVR/floor-capture identity. Conjecture 5 states a directional inversion of the standard LVR framing of Milionis et al. [2024]: for the protocol-owned LP position, LVR-equivalent value is captured into the floor rather than lost to arbitrageurs. Promoting this to a quantitative theorem requires a unit reconciliation between LVR’s per-unit-time stochastic-calculus formulation and M^2 ’s discrete event-driven floor framing, together with a careful per-leg accounting of the asymmetric fee. We have not completed this reconciliation; the directional content is sufficient for the qualitative arguments in Sections 2.3 and 4.6, but the quantitative identity is left open.

(P7) Sequenced bank-run strategies. Theorem 6 bounds the attacker’s proceeds within the *simultaneous*-strategy class, in which the dump and the redemption occur in a single block without an intervening `collectFees` realization. A sequenced strategy that interleaves LP dumps, `collectFees` calls, and redemptions within or across blocks can extract additional value scaling with the attacker’s residual holdings (Section 5.2, “Sequenced strategies”). Characterizing the supremum of attacker proceeds across the full sequenced-strategy class—which requires modeling the equilibrium response of the `collectFees`-bounty payer and the v4 fee-realization timing dynamics, as well as the cross-block adversary’s interaction with the (P4) builder-relay-collusion class—is left open.

8.6 Regulatory framing for academic readers

We close with a brief academic positioning on the regulatory question. This is not legal advice; any deployer must obtain qualified counsel in every relevant jurisdiction (operator domicile, target user geographies, listing venues). Our purpose here is to map M^2 's on-chain properties to the dimensions along which existing securities and digital-asset frameworks evaluate tokens.

Securities-style frameworks. Howey-style tests (U.S. securities law), the Markets in Crypto-Assets (MiCA) regulation (EU), the Virtual Asset Regulatory Authority (VARA) rules (Dubai), and the Monetary Authority of Singapore (MAS) Payment Services Act each ask different versions of the question “does this asset represent an investment contract.” M^2 has features that weigh in both directions: the redemption-guarantee and the operator’s affirmative duty to route revenue could be read as a fixed-return commitment (weighing toward securities classification); the absence of any pooled-investment management vehicle, the floating spot, the immutability of the parameters, and the protocol’s structural independence from the operator’s post-genesis discretion weigh against. We take no position on the outcome in any jurisdiction. Counsel will produce a different answer in each.

Spectrum positioning. On the canonical asset spectrum (utility token—governance token—asset-backed—e-money—security), M^2 is closest to “asset-backed” in the sense of an asset with a contractual claim on a reserve. It is explicitly not an e-money token under MiCA (no peg, no redemption-at-par); it is not a pure utility token (the floor is the central feature); it has no governance, so it is not a governance token. The closest TradFi categorization is “asset-backed warrant or note” or, in some jurisdictions, a redeemable preferred-share analog—each of which is treated differently in different regimes.

Jurisdictional reach. Smart contracts are address-agnostic and globally reachable. Geofencing operates at the front-end (user-interface) layer, not at the contract layer. A deployer who wishes to restrict access in specific jurisdictions can do so only through interface-level filters, and these can be bypassed by direct contract interaction. This is a feature of the substrate, not of M^2 specifically, and the legal exposure it produces should be evaluated by counsel for every target jurisdiction.

Recommendation. Deployers should obtain counsel before launch. Where the public materials describing the protocol use the language of “guaranteed floor” or “redemption,” counsel should review them; while these are accurate technical descriptions of contract behavior, regulators across jurisdictions read them as making investment-contract claims. This paper makes the technical claims; the legal characterization is a separate exercise.

9 Conclusion

We have specified, analyzed, and characterized M^2 , a tokenomic primitive that combines a fixed-supply redemption-backed token, an asymmetric-fee Uniswap-v4 hook, and a hardcoded revenue router to enforce a contractually monotone non-decreasing redemption floor. Our three contributions, stated precisely in Section 1, are sharper at the close. (C1) The primitive is fully specified: four immutable contracts, four free design parameters (revenue rate, LP seed fraction, asymmetric fee, treasury seed), a genesis ceremony that enforces $F_0 = P_0$, and a complete operation set Ops over which the invariants are claimed. (C2) The floor monotonicity property is proved by case analysis over Ops (Theorem 2), the integer-arithmetic redemption residual is shown to accumulate strictly in the treasury’s favor (Lemma 1), the redemption-fairness corollary inverts the classical [Chen et al. \[2010\]](#) first-mover-advantage result (Theorem 4), and

the composite-strategy bank-run game yields a state-only saturation quantity A^* above which the adversarial premium over pure redemption is a bounded constant (Theorem 6). (C3) The protocol’s operating envelope is characterized through a two-track simulator stack, with cross-track agreement enforced as a paper-stopping invariant and a reproducibility artifact deposited at Zenodo DOI [10.5281/zenodo.20255141](https://doi.org/10.5281/zenodo.20255141).

The cross-domain framing we have adopted positions M^2 as the on-chain implementation of three TradFi capital-return instruments that do not coexist in any single offering we are aware of: the closed-end fund with NAV redemption (and our addition: monotone NAV); the perpetual sponsor-funded buyback program (with deterministic, no-announcement-gap execution and burn-not-retire disposition); and the Pigovian sell tax recycled into common-pool value rather than third-party fisc. The composition of these three instruments is the substantive design contribution; the formal-methods apparatus and the quantitative-envelope characterization are the technical contributions that support the design’s defensibility.

What we do not claim, restated. The work proves a structural property of the protocol’s operation set. It does not eliminate smart-contract residual risk, does not promise robustness to the backing stablecoin’s issuer freezing the treasury, does not address operator fraud at the revenue-deposit layer, does not predict token prices, and does not offer regulatory classification. Each of these residual risks is addressed in Section 8 and is properly handled by deployment-time audit, counsel, and disclosure rather than by the contract code.

Future work. Three threads warrant follow-up. First, an *empirical* paper: after live deployment, an event-study analysis of the floor’s response to revenue shocks, the redemption-rate distribution, and the realized run-resistance under genuine adversarial pressure. Such a paper would target a top finance journal (*Journal of Financial Markets*, *Review of Financial Studies*) and would supply the empirical content that the present mechanism-design paper cannot. Second, a *mechanized-verification* paper formalizing Theorem 2 against the deployed bytecode under Halmos and Certora, closing open problem (P5) of Section 8. Third, *variants*: multi-stable backing (each stable a separate token deployment; the user diversifies across instances rather than across reserves within a single instance), operator-bonding designs (the operator’s revenue-deposit obligation is collateralized at genesis), and dynamic-fee schedules whose f_s adjusts to the spot-floor gap. Each variant is a separate token, by the immutability discipline of the design; comparing them empirically is part of the second thread above.

The narrower technical observation we leave the reader with: the on-chain substrate makes possible a primitive—a token with a contractually-monotone-non-decreasing per-unit reserve claim—that TradFi capital-returns legal infrastructure cannot enforce. The substantive observation is that the design space for such primitives has been substantially under-explored, that the structural failure modes of prior on-chain treasury-backed tokens are not failures of the on-chain substrate but of designs that retained discretionary surface area (governance, treasury composition, post-genesis minting), and that the M^2 design discipline—no discretion, no upgrade, no governance, no post-genesis minting, single transparent stable, one-way treasury inflow—is the load-bearing piece that converts a rhetorical claim into a proved theorem.

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Conflict of interest. The author intends to deploy a reference implementation of the M^2 protocol described in this paper. The mechanism is intended for open-source release as a non-proprietary tokenomics primitive available to any DeFi project; the author claims no proprietary rights over the design and does not hold a financial interest in any specific instance of the protocol beyond the reference deployment. Readers should weigh this intended deployment when interpreting the qualitative framing of the paper, though all quantitative claims are derivable from the protocol specification and the closed-form analysis of Sections 4 and 5 and do not depend on any operational assumption about a specific deployment.

Code and data availability. The simulator code, figure-generation scripts, and reference contract implementation will be released under an open-source license (license to be finalized at preprint posting) and deposited at the Zenodo DOI referenced in the abstract.

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A Symbol Table

For quick reference, Table 5 collects the notation used throughout the paper. Symbols introduced only locally (e.g. in a single proof) are not repeated here. Subscripts $(\cdot)_0$ denote genesis values; subscripts $(\cdot)_{\text{old}}/(\cdot)_{\text{new}}$ denote pre/post-operation values; superscript $(\cdot)^*$ denotes an equilibrium or optimum.

B Formula Reference

For quick reference, the load-bearing formulas of Sections 3–5 are collected below. Cross-references to the source equations and theorems are given in parentheses; this appendix is a digest, not a substitute.

Floor and spot prices.

$$F = T/S \quad (\text{Section 3.2, Eq. (1)})$$

$$P = L_s/L_t \quad (\text{Section 3.4})$$

Redemption payout for N tokens.

$$P = \lfloor N \cdot T/S \rfloor, \quad r = (N \cdot T) \bmod S \quad (\text{Lemma 1, Eq. (10)})$$

Post-redemption: $T' = T - P$, $S' = S - N$, with $F'/F - 1 = r/(T \cdot S')$ on the integer-arithmetic residual.

LP swap, sell side (with hook fee f_s on token input N_{in}).

$$\begin{aligned} \text{Stable out: } \Delta L_s &= L_s - k/(L_t + (1 - f_s)N_{\text{in}}) \\ \text{Fee accrual: } \Phi_t &\rightarrow \Phi_t + f_s N_{\text{in}} \quad (\text{Section 4.3, Case 5}) \end{aligned}$$

LP swap, buy side (with hook fee f_b on stable input X_{in}).

$$\text{Tokens out: } \Delta L_t = L_t - k/(L_s + (1 - f_b)X_{\text{in}}); \quad \Phi_s \rightarrow \Phi_s + f_b X_{\text{in}}$$

collectFees distribution.

$$K_b = \lfloor 0.0025K \rfloor, \quad U_b = \lfloor 0.0025U \rfloor, \quad K_{\text{burn}} = K - K_b, \quad U_{\text{treas}} = U - U_b$$

(Section 3.6, Section 4.3 Case 7).

LP half-life under deterministic monthly buy size B .

$$n_{1/2} = L_{s,0}/B \quad (\text{Theorem 8, Eq. (22)})$$

Bank-run saturation quantity.

$$A^* = \frac{1}{1 - f_s} \left(\sqrt{\frac{k(1 - f_s)}{F}} - L_t \right) \quad (\text{Theorem 6, Eq. (18)})$$

Bank-run premium (state-only constant, $N > A^*$).

$$\begin{aligned} \Delta^* &= \Pi^*(N) - NF = \left[L_s - \sqrt{kF/(1 - f_s)} \right] - A^*F \\ &= \left(\sqrt{L_s} - \sqrt{L_t F/(1 - f_s)} \right)^2 \\ &= L_s \cdot \left(1 - \sqrt{F/((1 - f_s)P)} \right)^2 \end{aligned}$$

(Theorem 6, Eqs. (20), (21)).

collectFees calling equilibrium under atomless competition.

$$\Delta t^* = g/(0.0025 \beta) \quad (\text{Theorem 9, Eq. (23)})$$

MEV leakage upper bound on protocol buy-and-burn.

$$\mathbb{E}[\text{MEV}] \leq (1 - p) \cdot \varepsilon_{\text{pub}}(B/N_{\text{TWAP}}, L_s) \cdot B + p \cdot \varepsilon_{\text{priv}} \cdot B \quad (\text{Theorem 10, Eq. (24)})$$

Genesis floor-spot constraint.

$$T_0/S_0 = L_{s,0}/L_{t,0} \quad (\text{forces } F_0 = P_0; \text{ Section 3.7, Eq. (7)})$$

C With-Fees vs. Fee-Free Curve-Math Corrections

This appendix tabulates the $O(f_b) = O(10^{-3})$ corrections to the baseline projection of Section 6.2 when the 0.10% buy fee is taken into account on each `BuyAndBurn` (Case 2 of Theorem 2). Under fee-free curve math each 50/50-split buy of size $B = R/2$ contributes B to the LP's stable reserve and burns $Y(B) = L_t - k/(L_s + B)$ tokens; under with-fees mode the effective input to the curve is $(1 - f_b)B$, the burned amount is $Y_{\text{wf}}(B) = L_t - k/(L_s + (1 - f_b)B)$, and the fee residue $f_b B$ accrues to Φ_s for realization to the treasury on the next `collectFees` call. The two modes converge to within the 0.25%-per-side caller bounty after collect-realization. A full reproducibility tabulation across the 36-month baseline horizon, alongside the with-fees Track A traces and the Track B fuzz state-diffs from the agreement gate of Section 6.1, will accompany the simulator artifact at the Zenodo DOI cited in the abstract; the appendix at submission time will include the resulting tables.

Table 5: Notation used throughout the paper.

Symbol	Meaning
T	Treasury stablecoin balance, in the stable’s smallest unit
S	Total token supply (LP-locked + vesting + circulating), in 10^{-18} -token units
L_t, L_s	Active LP reserves (token, stable) on the protocol-owned v4 position
$F = T/S$	Floor (redemption) price, returned as 18-decimal fixed-point regardless of the stable’s native decimals
$P = L_s/L_t$	AMM spot price
$k = L_t L_s$	Constant-product invariant of the LP curve
Φ_t, Φ_s	Raw unrealized fee mass on the LP position (token, stable), awaiting realization to the protocol on the next <code>collectFees</code> call; v4 stores this as a Q128.128 accumulator (Section 4.1)
K, U	Realized fee mass at <code>collectFees</code> , $(K, U) = (\Phi_t, \Phi_s)$ at the moment of realization
$K_{\text{burn}}, U_{\text{treas}}$	Net burned tokens / net stable to treasury after the 0.25% caller bounty
K_b, U_b	Caller bounty paid to the <code>collectFees</code> invoker (token, stable)
f_b, f_s	Hook fees on buys (0.10%) and sells (3%)
N	Token quantity held by an attacker or holder
N_{in}	Token input to an LP sell swap
X_{in}	Stable input to an LP buy swap
B	Monthly buy-and-burn size in stable ($B = R/2$ at canonical 50/50 split)
R	Monthly revenue routed through <code>routeRevenue</code>
A^*	Saturation quantity in the bank-run analysis (Theorem 6); state-only constant
Δ^*	Run premium over pure redemption; state-only constant (Theorem 6)
N^{cross}	Naive pure-strategy crossover where pure dump and pure redemption yield equal proceeds (Section 5.2)
$n_{1/2}$	LP token-side half-life (Theorem 8); $n_{1/2} = L_{s,0}/B$
Δt^*	Equilibrium inter-call interval for <code>collectFees</code> (Theorem 9)
β	Per-unit-time fee accumulation rate (Theorem 9)
g	Gas-plus-search cost of a <code>collectFees</code> call
$Q = 10^{18}$	18-decimal scaling factor
$d_s, Q_s = 10^{d_s}$	Backing stable’s decimal count and smallest-unit scale
$\llbracket \cdot \rrbracket$	Contract-faithful integer evaluation (vs. the abstract rational)
n	Discrete month index since genesis
t	Continuous time (seconds since genesis)
Ops	Protocol operation set (Section 4.1, Eq. (9))